

WEEK 2 REPORT

SUB-GROUP -14

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Team Charter

SLU 06007 DVT Team 14 Team Charter

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Team Lead	Soumojit Maitra (soumojitcloud@gmail.com)
Team Members Roles and Responsibilities	1. Dataset Overview, Structure & Data Profiling – Siddhi & Khyati 2. Visualizations (4 Charts/Visuals) – Nial & Laxmi 3. Exploratory Data Analysis (EDA) & Visualization – Soumojit & Sam 4. Key Findings, Business Insights & Recommendations – Murali & Paul 5. Final EDA Report Compilation, Review & Validation – Soumojit
Mission, Vision Objectives & Core Values	Mission: To work together to deliver practical, high-quality recommendations that meet our sponsor's needs and can be implemented

	successfully. Vision Objectives: Success means working collaboratively, communicating openly, meeting deadlines, and delivering a valuable outcome for our sponsor. Core Values: • Integrity: We act honestly, ethically, and transparently in all our work. • Accountability: We take responsibility for our tasks and commitments. • Discipline: We stay organized, meet deadlines, and maintain high standards. • Respect: We value each other's ideas, communicate professionally, and embrace diversity. • Innovation: We encourage creative thinking and practical solutions to deliver value.
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Internal Checks, Balances, and Reviews	Team Accountability and Performance
Accountability Mechanics	• Each team member is responsible for completing their assigned tasks on time, communicating progress, and asking for support when needed. • Sub-teams will provide brief progress updates before each full team meeting and notify the team of any challenges that may affect deadlines. • The full team will meet once a week (or more frequently if needed) to review progress, discuss challenges, assign new tasks, and ensure everyone is aligned with the project goals. • Individual performance will be measured using SMART goals, ensuring tasks are:

	- Specific: Clearly defined responsibilities. - Measurable: Progress tracked through completed deliverables. - Achievable: Tasks assigned according to each member's skills and capacity. - Realistic: Deadlines are practical and agreed upon by the team. - Time-based: All tasks have clear due dates.
Operations: Assignments	Investigate trends within Excelerate's Opportunity dataset by examining applications, outreach channels, locations, scholarship categories, and engagement patterns. Use charts, tables, and summary statistics to identify key trends, compare different groups, and discover relationships within the data.
Meetings	Team members met via Google Meet on 16 July 2026. We discussed week 2 deliverables to make sure everyone understood our tasks and responsibilities, and at the end we were assigned to our tasks and responsibilities.
Communication Guidelines & Status Updates	Communication Guidelines: The team lead made sure that every team member knew and understood their task; he shared updates and communicated mainly on Google Chat. Status Updates: Team members shared the project through Google Chat, and we used Google Docs for easy access to view and update the documents.
Deadlines	Project Draft is due July 19, The final project is due by July 20.

Dataset Overview, Structure and Data Profiling

1. Executive Summary

This report profiles the final analysis-ready dataset (**final_cleaned_dataset.xlsx**), documenting its dimensions, structure, field types, and data quality profile exactly as they exist in the uploaded file. It validates the dataset against four key data quality dimensions—**completeness, consistency, validity, and uniqueness**—with detailed findings presented in Sections 8 and 12. No additional cleaning, preprocessing, feature engineering, or exploratory data analysis has been performed within this report; all statistics and observations are derived directly from the uploaded dataset.

The dataset contains **5,730 records** across **20 fields**, with a fully unique and complete primary key and no fully duplicated rows. Numeric fields fall within materially tighter, more plausible ranges than in the earlier cleaning stage, and administrative Boolean flags are stored as clean numeric values. A small number of residual anomalies remain and are documented for transparency, including concentrated missing values in a few fields, **11 records** containing embedded text fragments, and **one** anomalous epoch-date value. These issues are minor and do not prevent the dataset from being used for subsequent exploratory data analysis (EDA), visualization, and business insight generation.

2. Dataset Purpose and Business Context

Each record in the dataset represents a single opportunity available on the source learning platform — classified into **10 primary category values**, led by **Internship (1,591 records, 27.8%)** and **Career (1,087 records, 19.0%)**, with smaller categories including **Competition, Course, Event, Engagement, Masterclass, Job Simulation, Program, and XploreU**.

The **20 retained fields** describe four business dimensions of each opportunity: **identification and classification** (*opportunity_id, code, category, role, location*), **commercial and scheduling terms** (*fee, currency_type, microscholarship, duration, duration_type, last_date_to_apply*), **descriptive and marketing content** (*name, short_description, long_description, role_responsibility, image_link*), and **administrative metadata** (*created_at, modified_at, is_archived, is_auto_approve*).

This structure supports catalog management and reporting on opportunities available to learners, though—as detailed in **Section 12**—several fields (particularly **is_archived, role_responsibility, and role**) carry substantial missingness that constrains certain business reporting use cases.

The business impact of this missingness is concrete: with **role** unpopulated for **67.0%** of records and **role_responsibility** unpopulated for **81.2%**, any dashboard or report intended to break down opportunities by role, or to summarize expected responsibilities for prospective applicants, will only reflect a minority of the catalog. Qualitative or role-based reporting should be treated as directional rather than comprehensive until these fields are more fully populated at the source.

3. Dataset Source

The dataset was supplied as a single Microsoft Excel workbook (**final_cleaned_dataset.xlsx**) containing one worksheet (**Sheet1**). The **opportunity_id** field follows the pattern "**Opportunity#**" followed by a unique alphanumeric identifier, and the **code** field contains short alphanumeric business codes (e.g., "**M636023**"), consistent with an extract from a platform's **Opportunity** entity.

No metadata describing the extraction date, refresh frequency, or export/transformation process accompanies this file. This section reports only what is directly observable in the file itself. The absence of this metadata has a direct reproducibility implication: without a recorded extraction date or refresh cadence, it is not possible to confirm whether this file represents a point-in-time snapshot or a rolling export, or to reliably reproduce an equivalent extract at a later date for comparison.

4. Dataset Overview

The dataset is tabular, with each of the **5,730 rows** corresponding to one opportunity record and each of the **20 columns** corresponding to one attribute of that opportunity. All records share the same schema; there is no nesting or multi-sheet structure in the uploaded file.

Table 4.1 — Headline characteristics of the uploaded final cleaned dataset.

Attribute	Value
Records (rows)	5,730
Fields (columns)	20
Worksheets	1 (Sheet1)
File format	Microsoft Excel Workbook (.xlsx)
Primary identifier	opportunity_id (100% unique, 100% complete)

5. Dataset Dimensions

Table 5.1 — Dataset dimensions and storage characteristics of the final cleaned dataset.

Metric	Value
Total records	5,730
Total fields	20
Total cells	114,600

Worksheets	1
File size on disk	0.821 MB (861,334 bytes)
In-memory size (deep)	7.18 MB (7,528,097 bytes)

For context on how the schema has evolved, the original raw export of this dataset (documented separately in the **Week 1 Dataset Overview**) contained **33 columns**, including **nine nested JSON relationship fields** (*Badge*, *CareerAddOn*, *Cohort*, *Eligibility*, *Reward*, *Panellist*, *Testimonial*, *DropoutTransaction*, and *NotStartedTransaction*) together with several highly sparse or low-relevance fields. The **20-column schema** profiled in this report excludes these JSON relationship fields, reflecting a deliberate refinement toward scalar, directly analyzable attributes that are more suitable for reporting, visualization, and exploratory data analysis.

6. Dataset Structure

Every row represents **one opportunity record**, uniquely identified by **opportunity_id** (**0 missing, 0 duplicates**). No fully duplicated rows exist anywhere in the dataset. One duplicate value exists in the secondary **code** field, where **2 of the 5,730 records** share the same code value.

Table 6.1 — Structural integrity checks for the final cleaned dataset.

Structural Check	Result
Fully duplicated rows	0
Duplicate opportunity_id values	0
Missing opportunity_id values	0
Duplicate code values	1 pair (2 records)
Records with embedded form-schema text fragments	11 (0.19%) — see Section 12.5

7. Column Organization

The 20 fields group into six functional categories:

Table 7.1 — Column Organization by Functional Group

Table 7.1 — The 20 fields grouped according to their primary functional role within the cleaned Opportunity dataset.

Group	Fields	Count
Identifiers	opportunity_id, code	2
Categorical	category, role, location, duration_type, currency_type	5
Numerical	duration, fee, microscholarship	3
Boolean	is_archived, is_auto_approve	2
DateTime	created_at, modified_at, last_date_to_apply	3
Text / Media	name, short_description, long_description, role_responsibility, image_link	5

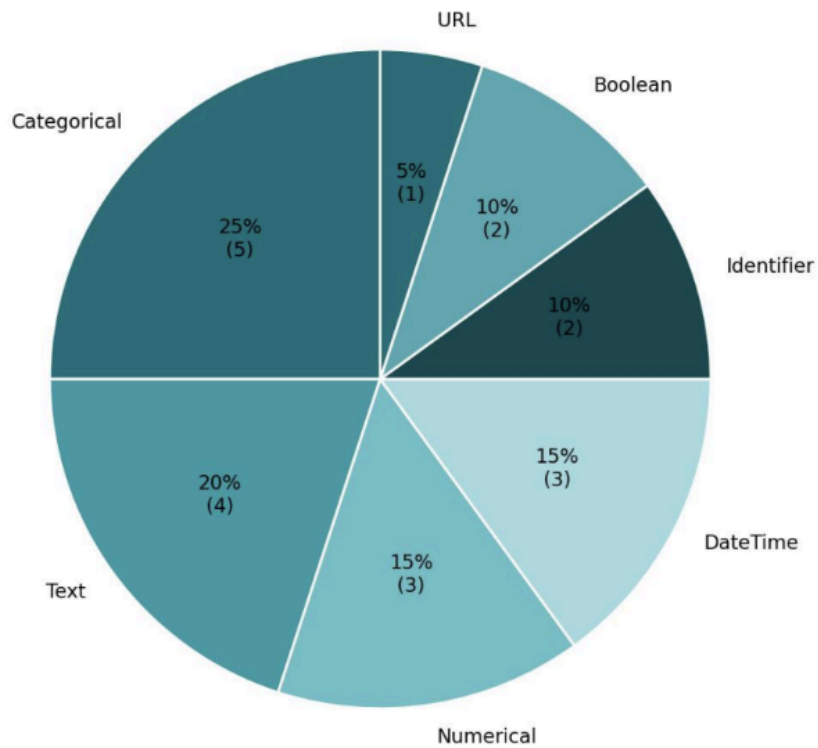


Figure 1. Feature category breakdown across all 20 columns. Text and Categorical fields make up half the schema (5 fields each); Identifier, Numerical, and DateTime fields (2–3 each) complete the structural backbone, alongside one URL field.

8. Dataset Schema

Table 8.1 presents the complete column-level schema, including stored data types, missing values, and unique value counts, as read directly from the uploaded file.

Table 8.1 — Dataset Schema

Table 8.1 — Complete column-level schema of the cleaned Opportunity dataset, including data types, missing values, and unique value counts.

Column	Dtype	Missing (n)	Missing (%)	Unique (n)
opportunity_id	object (string)	0	0.0	5,730
category	object (string)	3	0.1	10
code	object (string)	1	0.0	5,728
created_at	datetime64[us]	4	0.1	13

currency_type	object (string)	343	6.0	3
duration	float64	9	0.2	102
duration_type	object (string)	5	0.1	6
fee	float64	4	0.1	101
image_link	object	6	0.1	3,770
is_archived	float64	5,205	90.8	2
is_auto_approve	float64	3	0.1	2
last_date_to_apply	datetime64[us]	4	0.1	19
location	object (string)	1,455	25.4	3
long_description	object (string)	344	6.0	894
microscholarship	float64	5	0.1	157
modified_at	datetime64[us]	4	0.1	11
name	object	0	0.0	5,570
role	object (string)	3,837	67.0	429
role_responsibility	object (string)	4,651	81.2	88
short_description	object (string)	0	0.0	838

Missingness impact, by field: *is_archived* is missing for 90.8% of records, meaning archive status cannot be reliably reported on for the large majority of the catalog—absence of a value should not be interpreted as "active." *role_responsibility* (81.2% missing) and *role* (67.0% missing) similarly limit role-based reporting, as noted in Section 2. *location* (25.4% missing) constrains geographic or delivery-mode reporting, and *currency_type* (6.0% missing) limits full currency-aware coverage of the *fee* field. These four fields should be treated as directional indicators rather than complete records until source-side completeness improves.

9. Data Types and Feature Categories

No JSON-type (semi-structured object) fields remain in this dataset as a distinct data type; all 20 fields are stored as scalar text, numeric, or datetime values. Eleven records contain residual embedded JSON-like text fragments within otherwise scalar text fields (see Section 12.5), but this does not constitute a structural JSON field type in the current schema.

Table 9.1 — Feature Categories of the Dataset

Table 9.1 — Feature categories present in the cleaned dataset and the fields belonging to each category.

Feature Category	Field Count	Fields
Identifier	2	opportunity_id, code
Categorical	5	category, role, location, duration_type, currency_type
Numerical	3	duration, fee, microscholarship
Boolean	2	is_archived, is_auto_approve
DateTime	3	created_at, modified_at, last_date_to_apply
Text	4	name, short_description, long_description, role_responsibility
URL	1	image_link

10. Key Fields and Their Significance

Table 10.1 — Key Fields and Their Significance

Table 10.1 — Key fields that uniquely identify, classify, schedule, and manage opportunity records in the cleaned dataset.

Field	Significance
opportunity_id	Primary key. Fully unique (5,730/5,730) and fully populated; the canonical reference for every record.
code	Secondary business identifier; 99.98% unique, with one duplicate pair remaining.
category	Primary classification field; determines which of the 10 opportunity types a record belongs to and shapes how other fields (e.g., fee and duration) are typically interpreted.
created_at / modified_at / last_date_to_apply	The three structural timestamps record when a listing was created, last edited, and when it stops accepting applications.

is_archived / is_auto_approve	Administrative flags governing catalog visibility and application-approval workflow; stored as clean numeric Boolean values (0/1) in the cleaned dataset.
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11. Variable Classification

11.1 Numerical

duration, **fee**, and **microscholarship** are stored as **float64**. Ranges are materially narrower in this file than in a prior-stage cleaned version: the maximum **duration** is **359**, the maximum **fee** is **200,000**, and the maximum **microscholarship** is **50,000**.

11.2 Categorical

category (10 values), **role** (429 distinct values—largely free text rather than a fixed label set), **location** (3 values: *Work From Home*, *Virtual*, and *Kolkata*, plus **25.4%** missing), **duration_type** (6 values: *weeks*, *months*, *hours*, *days*, *minutes*, and *years*), and **currency_type** (3 values: *USD*, *INR*, and *EUR*).

11.3 Boolean

is_archived (values **0.0 / 1.0 / missing**—**90.8%** missing) and **is_auto_approve** (values **0.0 / 1.0 / missing**—**0.1%** missing) are both stored as clean two-state numeric fields in this file, with no non-conforming values detected.

11.4 DateTime

created_at, **modified_at**, and **last_date_to_apply** are all stored as native **datetime64[us]**. **created_at** spans **8 August 2022** to **28 May 2026**, while **last_date_to_apply** spans **1 January 1970** (a single anomalous value) to **28 December 2027**.

11.5 Text

name, **short_description**, **long_description**, and **role_responsibility** are free-form descriptive fields. **Eleven records** contain embedded form-schema JSON fragments within these fields (see **Section 12.5**).

11.6 URL

image_link stores the opportunity's cover image address. **3,770** of **5,730** values are unique (**65.8%**), indicating that some images are reused across multiple listings.

12. Comprehensive Data Profiling

12.1 Descriptive Statistics — Numerical Fields

Table 12.1 — Descriptive Statistics for Numerical Fields

Table 12.1 — Summary descriptive statistics for the numerical variables in the cleaned Opportunity dataset.

Field	Count	Mean	Std Dev	Min	25%	50% (Median)	75%	Max
duration	5,721	24.54	16.59	0.0 0	12.00	30.00	30.00	359.00
fee	5,726	57.45	2,792.1 2	0.0 0	0.00	0.00	0.00	200,000.0 0
microscholarship	5,725	175.6 6	832.73	0.0 0	100.0 0	120.00	120.0 0	50,000.00

The median (50%) is **0** for **fee**, reflecting that the majority of records carry no charge, while the mean of **57.45** is pulled upward by the minority of paid listings. **Duration** and **microscholarship** exhibit the same pattern, with medians well below their tail-driven means, although the difference is far less pronounced than in a prior-stage cleaned version of this dataset.

Outlier Note: Although the numerical ranges in this cleaned dataset are substantially narrower than those observed in an earlier cleaning stage, genuine outliers remain. The **fee** field reaches a maximum of **200,000** despite a median of **0**, while **microscholarship** reaches **50,000** despite a median of **120**. Because both variables are highly concentrated around zero or relatively small values, these upper-end observations disproportionately increase the mean compared with the median (fee: **mean = 57.45**, **median = 0**; microscholarship: **mean = 175.66**, **median = 120**). Consequently, analyses that rely on the **mean** rather than the **median** for these variables may overstate the typical fee or scholarship amount and should clearly acknowledge the influence of these high-value observations.

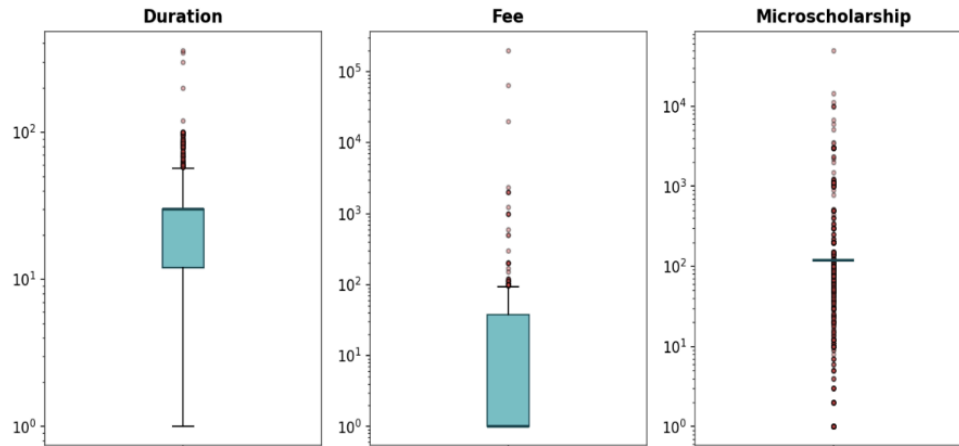


Figure 2. Box plots of **duration**, **fee**, and **microscholarship** (log scale, values greater than zero only). All three fields show a compact interquartile range with a moderate right-skewed tail of higher-value outliers, consistent with the descriptive statistics presented in **Table 12.1**.

12.2 Descriptive Statistics — Categorical Fields

Table 12.2 — Descriptive Statistics for Categorical Fields

Table 12.2 — Summary statistics for the principal categorical variables in the cleaned Opportunity dataset.

Field	Mode (Top Value)	Frequency	Distinct Values
category	Internship	1,591 (27.8%)	10
duration_type	weeks	4,286 (74.8%)	6
currency_type	USD	4,698 (87.3% of known)	3
location	Work From Home	2,356 (41.1%)	3
is_auto_approve	0.0 (False)	4,916 (85.8%)	2

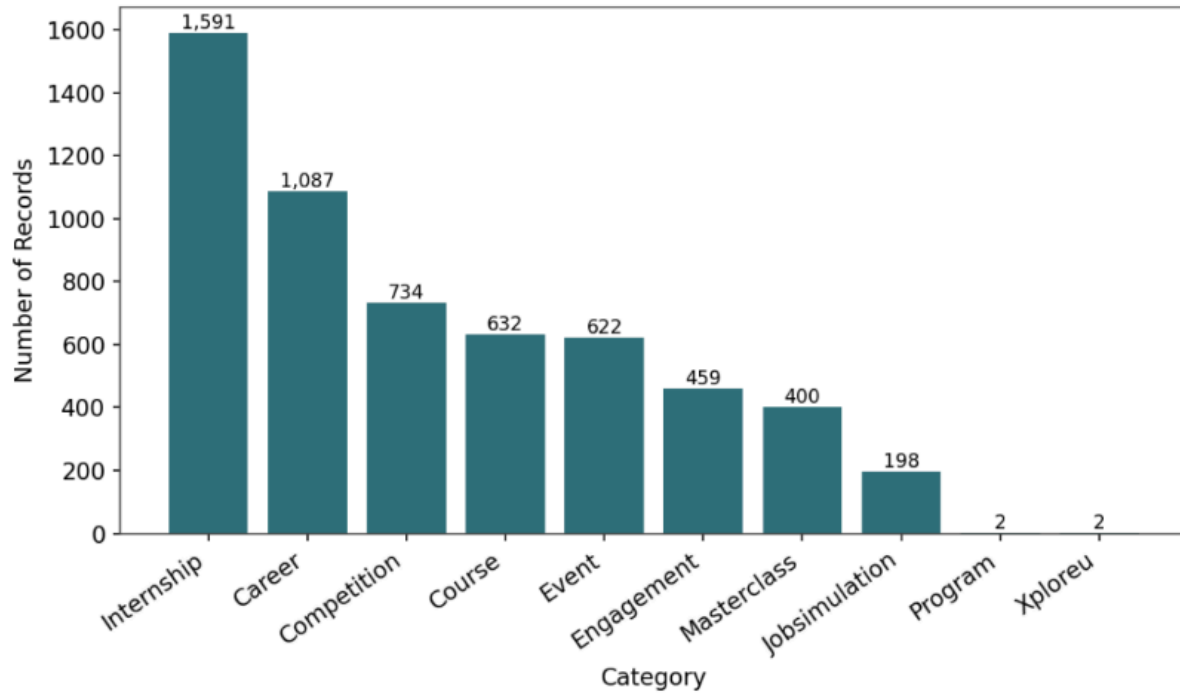


Figure 3. Record count by **category**. **Internship (1,591)** and **Career (1,087)** together account for **46.8%** of all records, while **Program** and **XploreU** are the smallest categories, with **2 records each**.

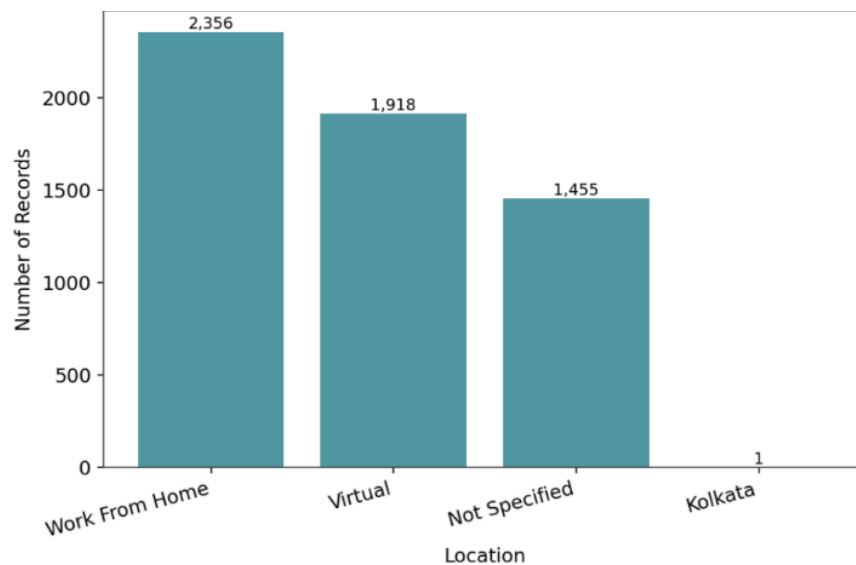


Figure 4. Record count by **location** value. **Work From Home (2,356)** and **Virtual (1,918)** account for the two populated location values; **1,455 records (25.4%)** have no location recorded, and a single record specifies "**Kolkata**" as a physical location.

12.3 Missing Values

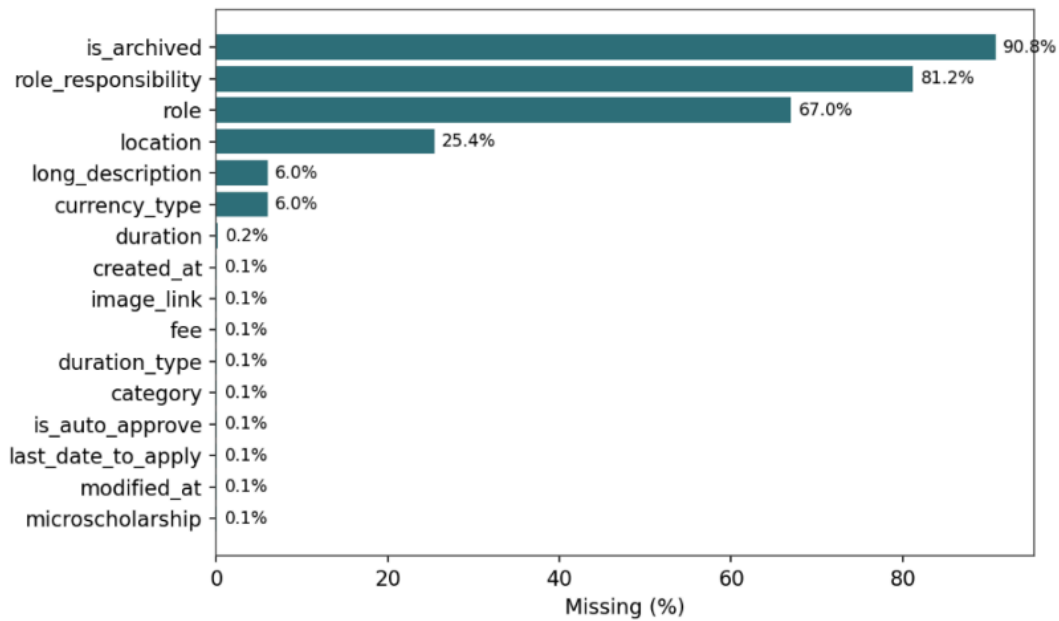


Figure 5. Missing value percentage for the seven fields with any missingness. **is_archived** is by far the most incomplete field (**90.8%**), followed by **role_responsibility** (**81.2%**) and **role** (**67.0%**). **location** shows a moderate level of missingness (**25.4%**), while **long_description**, **currency_type**, and **duration** each exhibit only minor missingness (**6.0% or below**).

The **13 fields** not shown in **Figure 5** (**opportunity_id**, **category**, **code**, **created_at**, **duration_type**, **fee**, **image_link**, **is_auto_approve**, **last_date_to_apply**, **microscholarship**, **modified_at**, **name**, and **short_description**) each have missingness at or below **0.1%**.

12.4 Uniqueness and Duplicates

- **Fully duplicated rows:** 0
- **Duplicate opportunity_id:** 0 (fully unique, fully populated primary key)
- **Duplicate code:** 1 pair (2 of 5,730 records share a code value with distinct opportunity_id values)
- **Highest-cardinality non-identifier field:** image_link (65.8% unique)
- **Lowest-cardinality fields:** is_archived, is_auto_approve, location, duration_type, currency_type (each under 0.5% unique), consistent with small fixed label sets.

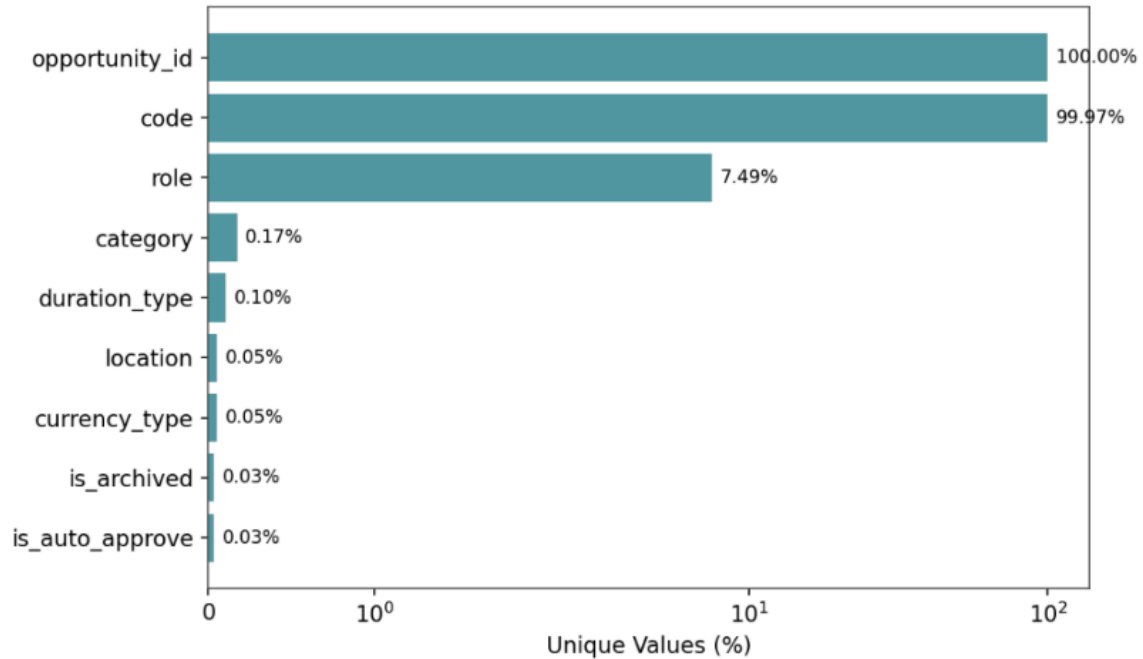


Figure 6. Cardinality (percentage of unique values) by field, log scale. **opportunity_id** is a pure identifier (100% unique); **image_link** and **name** show high but incomplete uniqueness (65.8% and 97.2%, respectively). Classification fields (**category**, **duration_type**, **currency_type**, and **location**) cluster near 0% unique, as expected for small fixed label sets.

12.5 Consistency and Validity

Eleven records (0.19% of the dataset) contain embedded, URL-encoded JSON-like text fragments (e.g., "%22label%22: %22...%22") inside one or more otherwise free-text fields, most commonly **image_link**, **long_description**, and **short_description**. Two of these eleven records also have corrupted **opportunity_id** and **code** values, matching a known source-export anomaly. The remaining nine records have valid identifiers but carry form-schema fragments embedded within their descriptive text fields, most extensively in two **long_description** values that contain what appears to be an entire nested application-form definition rather than opportunity description text.

In total, **11 records** are affected: **9** with valid identifiers carrying embedded text fragments, and **2** with corrupted identifier fields themselves. The full list of affected **opportunity_id** values has been omitted from this report for conciseness and can be provided separately on request.

These embedded fragments are source-level anomalies, not artifacts introduced during cleaning. The observed pattern (URL-encoded JSON application-form definitions appearing within free-text fields) is consistent with a source-system export issue rather than a transformation error, and no attempt was made to parse, repair, or remove this content as part of this profiling report.

Outside of these eleven records, **category** (10 values), **duration_type** (6 values), **currency_type** (3 values), and **location** (3 values) all exhibit internally consistent, non-overlapping label sets, with no

singular/plural duplication or casing inconsistencies detected. This reflects a marked improvement in standardization, consistent with a further-refined cleaning stage.

Validity Flag — Epoch Date Anomaly: One **last_date_to_apply** value is recorded as **1970-01-01**, the Unix epoch date conventionally produced when a null or zero timestamp is coerced into a date field. This value is treated as a validity issue rather than a legitimate historical deadline, as an opportunity with an application deadline of **1 January 1970** predates the platform itself. Accordingly, it should not be included in any deadline-based reporting or filtering without correction. The value is documented here for business review but has not been modified as part of this profiling report.

12.6 Completeness

Table 12.3 — Completeness by Field

Table 12.3 — Completeness tiers for all fields in the cleaned Opportunity dataset.

Completeness Tier	Field Count	Fields
≥ 99.8% complete	13	opportunity_id, category, code, created_at, duration_type, fee, image_link, is_auto_approve, last_date_to_apply, microscholarship, modified_at, name, short_description
94–99.8% complete	3	duration (99.8%), long_description (94.0%), currency_type (94.0%)
70–80% complete	1	location (74.6%)
Below complete 35%	2	role (33.0%), role_responsibility (18.8%)
Below complete 10%	1	is_archived (9.2%)

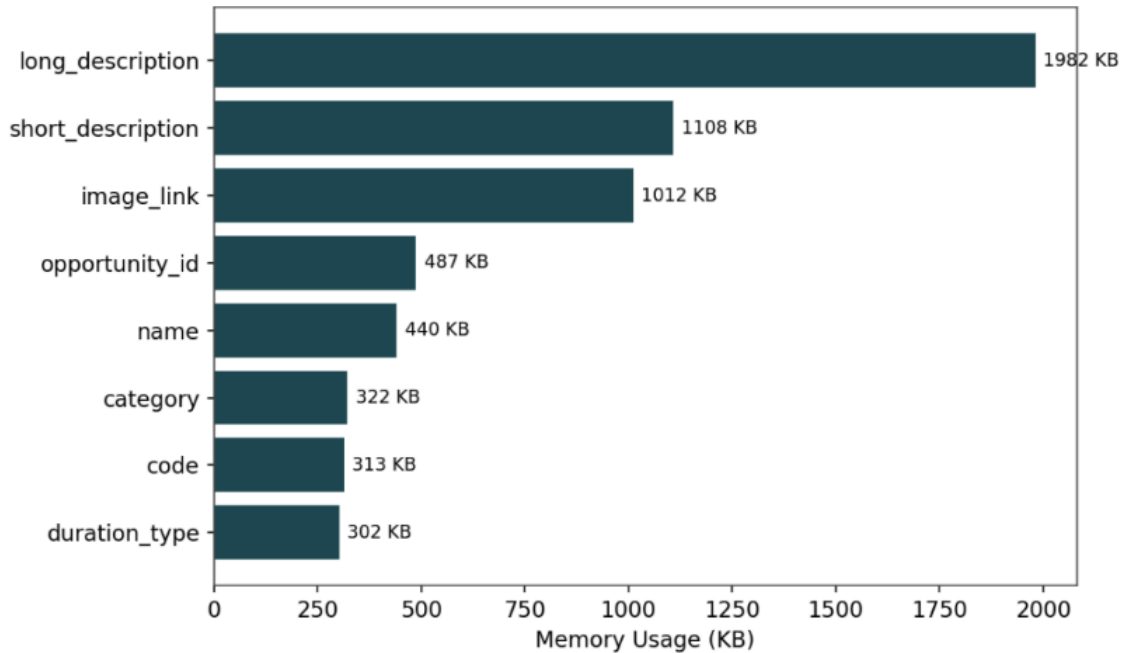


Figure 7. Top 8 fields by in-memory footprint. *long_description* and *short_description* dominate memory usage due to their free-text content, followed by *image_link*, reflecting content richness rather than a data quality issue.

13. Overall Data Quality Summary

Table 13.1 — Overall Data Quality Summary

Table 13.1 — Summary of the key data quality dimensions for the final cleaned Opportunity dataset.

Dimension	Finding
Uniqueness	Primary key (opportunity_id) is fully unique and complete; 1 duplicate code pair remains.
Completeness	13 of 20 fields are at or above 99.8% complete ; is_archived (9.2%) and role_responsibility (18.8%) are the least complete fields.
Consistency	category, duration_type, currency_type, and location each show clean, non-overlapping label sets; no singular/plural or casing variants detected in this file.
Validity	11 of 5,730 records (0.19%) contain embedded form-schema text fragments in one or more text fields; 2 of these also have corrupted identifier fields.

Accuracy (numeric)	duration, fee, and microscholarship fall within materially tighter ranges than a prior-stage cleaned version, with no records exceeding 359 (duration) , 200,000 (fee) , or 50,000 (microscholarship) .
Timeliness	Timestamp fields span August 2022 to beyond December 2027 , stored as native datetime values; one epoch-date anomaly (1970-01-01) remains in last_date_to_apply .

This report describes the dataset in its **final analysis-ready state**, with residual anomalies—including **high missingness in select fields, eleven embedded-fragment records, and one epoch-date value**—documented for transparency rather than resolved within this report. No additional cleaning, preprocessing, feature engineering, or exploratory data analysis has been performed within this report; all figures, tables, and statistics presented above are derived directly from **final_cleaned_dataset.xlsx** as uploaded.

14. Files Delivered

Table 14.1 — Files Delivered

File	Description
final_cleaned_dataset.xlsx	Source dataset profiled in this report (5,730 records × 20 fields), as uploaded.
Dataset Overview, Structure, and Data Profiling Report (this document)	Publication-quality profiling report covering dataset overview, schema, structure, data types, completeness, uniqueness, consistency, validity, descriptive statistics, and overall data quality assessment .

Note: No **Data Profiling Notebook (.ipynb)** has been produced alongside this report. If a notebook containing the underlying profiling code is required for reproducibility, auditability, or future maintenance, it can be prepared and provided as a separate deliverable upon request.

15. Conclusion

Based on this profiling pass, the **final_cleaned_dataset.xlsx** file is structurally sound, internally validated, and analysis-ready. The primary key is fully unique and complete, categorical fields exhibit clean and non-overlapping label sets, and numerical fields fall within materially tighter and more plausible ranges than those observed in the earlier cleaning stage.

The source-level anomalies identified in this report—namely concentrated missingness in **is_archived, role, and role_responsibility**; **11 records** containing embedded form-schema text fragments; and **one**

epoch-date value in last_date_to_apply—are documented for transparency and future business review. These issues are limited in scope and do not significantly affect the dataset's suitability for analysis or dashboard development. Instead, they should be regarded as known limitations and considered when interpreting analytical results.

This **Dataset Overview, Structure, and Data Profiling Report** provides the foundation for the subsequent **Exploratory Data Analysis (EDA)**, **data visualizations**, and **business insights** presented in the **Week 2 deliverables**.

EXPLORATORY DATA ANALYSIS (EDA) REPORT

EDA Implementation Notebook (Google Colab): [EDA_Opportunity_Dataset.ipynb](#)

(Implemented using Pandas and NumPy; visualizations are documented separately in the Visualization Report.)

Executive Overview

This report presents the consolidated exploratory data analysis (EDA) findings for the Opportunity Analysis-Ready dataset. It integrates structural, statistical, and data-quality observations with business-oriented interpretations to support decision-making on data governance, platform strategy, and downstream reporting. Each section follows a consistent Observation → Interpretation → Business Insight → Report-Ready Summary structure to ensure findings are both analytically rigorous and directly usable in stakeholder communication.

Section 1: Dataset Overview and Shape

Observation. The dataset was loaded successfully using pandas and contains 5,730 records across 20 columns. This scale and structure remained consistent between the initial load and the refreshed notebook validation.

Interpretation. Each row corresponds to a single opportunity listing (internship, career, competition, course, or event), while the 20 columns capture identity, classification, timing, cost, location, descriptive text, and role-related attributes. The dataset's breadth of columns provides multiple analytical lenses, though the feature count is modest relative to what would be needed for deeper behavioral or engagement analysis.

Business Insight. A sample of 5,730 records is large enough to support statistically meaningful pattern identification across categories, durations, and fee structures, without introducing excessive processing overhead. However, because the feature set is relatively narrow (20 columns), the dataset is better suited to descriptive and structural analysis than to predictive modeling of user behavior; any roadmap for behavioral analytics would require enriching the schema with engagement or interaction-level data.

Summary: *The analysis-ready dataset comprises 5,730 records and 20 columns, spanning opportunity type, category, duration, fee, location, and descriptive metadata. This sample size is sufficient to support platform-wide trend analysis, though the current schema depth is best suited to structural and descriptive reporting rather than behavioral analytics.*

Section 2: Column Information

Observation. The dataset contains four functional groups of columns: identifiers (opportunity_id, code, name), classification fields (category, currency_type, duration_type, location), numeric and date fields (duration, fee, microscholarship, created_at, modified_at, last_date_to_apply, is_archived, is_auto_approve), and free-text fields (long_description, short_description, role, role_responsibility).

Interpretation. The coexistence of structured and unstructured fields means the dataset can support both quantitative trend analysis and qualitative review of how opportunities are described and positioned to applicants.

Business Insight. This dual structure enables the analytics function to serve two distinct stakeholder needs simultaneously: operational reporting (via structured fields) and content-quality auditing (via free-text fields). Any future analytics roadmap should treat these as separate workstreams, since text-field quality issues (detailed in Section 12) should not be allowed to distort structured KPI reporting.

***Summary:** The 20 columns combine unique identifiers, categorical classifiers, numerical measures, timestamp fields, and free-text descriptions, enabling multi-dimensional analysis of opportunity characteristics, timing, and cost structure alongside qualitative review of description quality.*

Section 3: Dataset Structure (Head/Tail Preview)

Observation. A preview of the first and last five records shows opportunity_id values consistently prefixed with "Opportunity#" and followed by an alphanumeric code, with categories (Competition, Career, Internship, Event) represented at both the beginning and end of the file.

Interpretation. Categories appear interspersed rather than grouped, indicating the raw file is not sorted by category and cannot be relied upon for positional segmentation.

Business Insight. Any category-based reporting must use explicit grouping and aggregation logic rather than row position or file order, ensuring category-level metrics remain accurate regardless of how records were originally appended to the source system.

***Summary:** A preview of the first and last five records confirms a consistent identifier format and shows that opportunity categories are distributed throughout the file rather than sorted by type, reinforcing the need for explicit aggregation in category-level analysis.*

Section 4: Data Type Summary

Observation. Per df.info() and df.dtypes, the dataset contains 3 datetime64[ns] columns (created_at, last_date_to_apply, modified_at), 5 float64 columns (duration, fee, is_archived, is_auto_approve, microscholarship), and 12 object columns. Total in-memory footprint is approximately 895.4 KB.

Interpretation. Dates and numeric fields are already correctly typed, indicating that upstream cleaning has been applied prior to this analysis stage and that minimal additional type-conversion work is required before statistical processing.

Business Insight. Correct upstream typing reduces the risk of computational errors in downstream date arithmetic and numeric aggregation, and confirms that the "analysis-ready" designation is largely accurate at the schema level — though, as later sections show, correct typing does not guarantee content-level data quality.

***Summary:** The dataset comprises 3 datetime columns, 5 numeric columns, and 12 object-type columns, totaling approximately 895.4 KB in memory. Correct typing of date and numeric fields confirms the dataset was properly pre-processed at the schema level, though downstream sections identify content-level quality issues that persist despite this.*

Section 5: Missing Value Analysis

Observation. `is_archived` is missing in 5,205 records (90.84%), `role_responsibility` in 4,651 (81.17%), and `role` in 3,837 (66.96%). `location` is missing in 1,455 records (25.39%). `long_description` and `currency_type` show moderate gaps of 6.00% and 5.99% respectively. All remaining fields have missingness below 0.16%, and `opportunity_id`, `name`, and `short_description` are fully populated.

Interpretation. The three highest-missingness fields (`is_archived`, `role_responsibility`, `role`) are likely optional at data entry or relevant only to a subset of opportunity types such as internships or jobs. Location, missing in roughly one-quarter of records, is a meaningful gap for any geographic segmentation, while most other fields are reliable at near-full population.

Business Insight. Fields with missingness above 65% are unsuitable as core analytical dimensions and should be treated as supplementary rather than authoritative; using them for full-population KPIs (e.g., "percentage of archived opportunities") without qualification would materially misstate coverage. Location gaps limit — but do not eliminate — the reliability of geographic analysis; findings in this area should be explicitly scoped to "records with available location information" rather than generalized to the full dataset. Implementing mandatory-field validation for `role_responsibility` at data entry would materially improve the platform's ability to support role-based matching in future analysis cycles.

***Summary:** Missing-value analysis reveals substantial data gaps in three fields: `is_archived` (5,205 missing, 90.84%), `role_responsibility` (4,651 missing, 81.17%), and `role` (3,837 missing, 66.96%). Location is missing in 1,455 records (25.39%), while `long_description` and `currency_type` show moderate gaps of 6.00% and 5.99% respectively. All remaining fields have missingness below 0.16%, with `opportunity_id`, `name`, and `short_description` fully populated. These patterns indicate that role- and archival-related fields are not consistently captured across all opportunity types and should be treated as supplementary in downstream reporting, while location-based findings should be scoped to the subset of records with available data rather than generalized platform-wide.*

Section 6: Duplicate Analysis

Observation. The duplicate check (`df.duplicated().sum()`) returned 0 duplicate rows across the full dataset, consistent across both the original and refreshed notebook runs.

Interpretation. No fully duplicated records exist; every row differs by at least one field.

Business Insight. The absence of duplicate rows means reported counts (e.g., category frequencies, currency splits) can be trusted as reflecting distinct records rather than being inflated by repeated entries, and no row-level deduplication step is required prior to analysis.

***Summary:** The dataset contains 0 duplicate rows, confirming that all 5,730 records are unique at the row level and require no further deduplication prior to analysis.*

Section 7: Summary Statistics

Observation. `created_at` spans from August 8, 2022 to May 28, 2026 (mean: June 17, 2024; count 5,726). `duration` has a mean of 24.54 against a median of 30.00 (std 16.59, max 359.00). `fee` has a mean of 57.45 against a median of 0.00 (std 2,792.12, max 200,000.00). `is_archived`, populated in only 525 records, shows a mean of 0.73.

Interpretation. The `created_at` range indicates nearly four years of opportunity creation activity captured in the dataset. Duration's median (30.0) matching its mode signals a reasonably central distribution with a long right tail toward longer-duration opportunities. Fee's wide divergence between mean and median, combined with a standard deviation roughly 49 times the mean, confirms extreme right-skew driven by a small number of high-fee outliers; the maximum fee of 200,000 is either a genuine premium offering or a data-entry anomaly warranting verification. Where populated, `is_archived` records skew toward archived status (73%), though this figure covers only 9% of total records and should not be extrapolated platform-wide.

Business Insight. Because fee's mean (57.45) sits far above its median (0.00), the median is the more representative statistic for describing typical opportunity cost; any pricing-strategy narrative built on the mean alone would materially overstate what a typical applicant pays. Similarly, `is_archived` statistics should be reported with an explicit caveat on their limited coverage (525 of 5,730 records) rather than presented as a platform-wide archival rate.

***Summary:** Summary statistics show opportunity creation dates ranging from August 8, 2022 to May 28, 2026 (mean: June 17, 2024). Duration has a mean of 24.54 and a median of 30.00 (std 16.59, max 359). Fee shows a mean of 57.45 against a median of 0.00 (std 2,792.12, max 200,000.00), indicating that while most opportunities are free, a small subset carries very high fees that heavily skew the average. Where populated, `is_archived` (525 non-null values, 9.2% of the dataset) shows a mean of 0.73, a figure that should be interpreted as indicative only, given its limited coverage.*

Section 8: Unique Values Analysis

Observation. `opportunity_id` has 5,730 unique values, matching the row count exactly. `code` has 5,728 unique values against 5,730 rows (the refreshed notebook confirms one code value, IZ18WQO, is duplicated). `name` has 5,570 unique values against 5,730 rows. `category` (10), `currency_type` (3), `duration_type` (6), and `location` (3) show low cardinality consistent with categorical grouping fields.

Interpretation. opportunity_id functions as the dataset's sole confirmed primary key. Both code and name contain minor duplication and should not be relied upon as unique identifiers in joins or deduplication logic.

Business Insight. Any analysis, matching, or record-linkage logic should reference opportunity_id exclusively; using code or name as a join key risks silently collapsing or misattributing a small number of distinct records (2 rows via code, roughly 160 rows via name).

***Summary:** Cardinality analysis confirms opportunity_id as the dataset's true unique identifier, with 5,730 unique values matching the total record count. code shows 5,728 unique values and name shows 5,570 unique values against 5,730 total rows, indicating minor duplication in both fields. Categorical fields such as category (10), currency_type (3), duration_type (6), and location (3) exhibit low cardinality consistent with their role as grouping variables.*

Section 9: Categorical Feature Analysis

Observation. category (10 values): Internship 1,591; Career 1,087; Competition 734; Course 632; Event 622; Engagement 459; Masterclass 400; Jobsimulation 198; Program 2; Xploreu 2. currency_type: USD 4,698; INR 355; EUR 334. duration_type: weeks 4,286; months 513; hours 371; days 369; minutes 144; years 42. location (among the 4,275 non-null records): Work From Home 2,356; Virtual 1,918; Kolkata 1. role (429 unique values) shows heavy case-variant duplication, e.g., Manager 720 / manager 37, Role 284 / role 26, Tester 64 / tester 18. Of 525 non-null is_archived records, 384 (73.1%) are archived; of 5,727 non-null is_auto_approve records, 811 (14.2%) are auto-approved.

Interpretation. Internship, Career, and Competition together account for 3,412 of 5,730 records (59.5%), indicating the platform's core offering is oriented toward skills and career development rather than short-form engagement content. Among the 4,275 records with location data, remote-format entries (Work From Home and Virtual) account for 4,274 (99.98%), with only a single record reflecting a physical location; this pattern is strong within the available data but covers only 74.6% of the dataset, since location is missing in the remainder. The role field's inconsistent casing indicates a data-entry standardization gap that would distort any role-based aggregation if left uncorrected.

Business Insight. The concentration in career-development categories suggests platform strategy and marketing investment are reasonably well-aligned with actual listing volume, and further growth opportunity may exist in underrepresented categories (Program, Xploreu) if there is unmet demand. Among records with available location information, the data indicates a strongly remote-oriented delivery model; however, because location is missing for 25.4% of records, this should be described as an indication supported by the available data rather than a definitive platform-wide characteristic. The role field's casing inconsistencies (e.g., "Manager" vs. "manager") should be normalized before any role-based segmentation or workforce-demand analysis is undertaken, since combined variants would materially change reported counts (e.g., Manager + manager = 757 combined).

***Summary:** Category-level analysis shows Internship as the leading opportunity type (1,591 records, 27.8%), followed by Career (1,087, 19.0%) and Competition (734, 12.8%), while Program and Xploreu*

are minimal at 2 records each. Currency data is dominated by USD (4,698 records), with INR (355) and EUR (334) representing smaller shares. Duration is most commonly measured in weeks (4,286 records, 74.9% of non-null values). Among the 4,275 records with location data (74.6% of the dataset), Work From Home (2,356) and Virtual (1,918) account for 4,274 of 4,275 entries, indicating a strongly remote-oriented delivery model within the available data. The role field exhibits case-variant duplication (e.g., "Manager" 720 vs. "manager" 37) that should be standardized before role-based analysis. Of the 525 non-null is_archived records, 384 (73.1%) are archived, and of the 5,727 non-null is_auto_approve records, 811 (14.2%) are auto-approved.

Section 10: Numerical Feature Analysis

Observation. duration: mean 24.54, median 30.0, mode 30.0, std 16.59, skewness available via notebook skew() output. fee: mean 57.45, median 0.0, mode 0.0, std 2,792.12. microscholarship: mean 175.66, median 120.0, mode 120.0, std 832.73.

Interpretation. Duration's mean, median, and mode cluster closely, indicating a moderately balanced distribution suitable for direct averaging in reporting. Fee and microscholarship both show pronounced right-skew — fee far more severely (std nearly 49x the mean) than microscholarship (std under 5x the mean) — confirming that a small number of high-value records disproportionately influence both averages.

Business Insight. Duration can be reported using mean-based statistics with reasonable confidence. Fee and microscholarship cannot: any "average fee" or "average microscholarship" figure presented without the corresponding median risks materially overstating typical cost or incentive value to stakeholders such as prospective applicants or partner institutions. The near-standardized microscholarship mode (120.0) suggests a deliberate, templated incentive tier rather than case-by-case financial grants.

Summary: Numerical analysis confirms varying degrees of skew across the three core numeric fields. Duration has a mean of 24.54 against a median/mode of 30.0 (std 16.59), representing a reasonably balanced distribution. Fee shows a mean of 57.45 against a median of 0.0 (std 2,792.12), confirming that the majority of opportunities carry no fee while a small number of high-value outliers substantially inflate the average. Microscholarship follows a similar but less extreme pattern (mean 175.66 vs. median 120.0, std 832.73), consistent with a standardized low-value incentive model rather than high-value case-by-case grants.

Section 11: Date Analysis

Observation. Record counts by Creation Year: 2022 – 25; 2023 – 2,123; 2024 – 2,127; 2025 – 895; 2026 – 556 (partial year, through May 28, 2026).

Interpretation. Listing volume increased sharply from 2022 to 2023 and held at a similarly high level through 2024, before declining in 2025 and 2026. The 2022 figure (25 records) likely reflects a partial onboarding period rather than a full year of activity, and the 85x increase into 2023 is consistent with either a scaling phase or a data-migration event rather than organic growth alone.

Business Insight. The 2023–2024 period represents the dataset's peak listing-activity window (4,250 of 5,726 dated records, 74.2%) and should be treated as the primary baseline for year-over-year comparisons. The apparent decline in 2025–2026 volumes should not be read as a business slowdown without further investigation, since 2026 is a partial year (data through May 28 only) and the underlying cause of the 2023 surge (organic growth vs. migration) has not been independently confirmed from this dataset alone. Any growth narrative built on raw record counts should be framed as an indication supported by listing data rather than a confirmed measure of business expansion.

Summary: *Year-wise analysis of opportunity creation shows an increase from 25 records in 2022 to 2,123 in 2023, sustained at 2,127 in 2024, before declining to 895 in 2025 and 556 in 2026 (a partial year, as the dataset's latest recorded date is May 28, 2026). The years 2023 and 2024 together account for 4,250 of 5,726 dated records (74.2%), marking this period as the peak of listing activity within the dataset's observed timeframe. Because 2026 reflects only a partial year and the cause of the 2023 increase is not independently confirmed, this trend should be treated as indicative rather than a definitive measure of business growth.*

Section 12: Text Feature Analysis

Observation. short_description: mean length 148.68 characters (median 200, max 744). long_description: mean 318.98 (median 440, max 4,784). role_responsibility: mean 24.09 (median 21, max 536). The refreshed notebook's categorical overview additionally surfaces frequent placeholder or templated entries, including a single long_description template repeated 3,501 times, and generic entries such as "Long Description" (179), "Short Description" (118), "test" (60), "ok" (14), and "abcd" (12).

Interpretation. The clustering of short_description and long_description lengths near fixed values (200 and 440 characters respectively) suggests standardized templates rather than fully bespoke content, while the recurrence of minimum lengths of 1 character and generic placeholder text across all three fields points to a meaningful volume of low-effort or incomplete entries rather than genuine descriptive content.

Business Insight. Text-based or NLP-driven initiatives (e.g., automated tagging, recommendation matching, or applicant-facing search) built on these fields would inherit this quality issue and should not be deployed without a content-quality audit and filtering step. The heavy repetition of a single long_description template (3,501 of 5,386 non-null records, 65.0%) also suggests many opportunities may be under-differentiated in how they are presented to applicants, which is a content and marketing consideration distinct from the underlying data quality issue.

Summary: *Text length analysis shows short_description has a mean length of 148.68 characters (median 200, max 744), long_description has a mean of 318.98 characters (median 440, max 4,784), and role_responsibility has a mean of 24.09 characters (median 21, max 536). The recurrence of minimum lengths of 1 character, combined with a single long_description template accounting for 65.0% of non-null entries and frequent generic placeholder values (e.g., "test," "abcd," "Short*

Description"), indicates a notable proportion of records contain templated, incomplete, or placeholder text rather than fully differentiated descriptive content.

Section 13: Logical Consistency Checks

Observation. Three consistency checks were performed: (1) records with fee < 0 — 0 rows returned; (2) records with duration ≤ 0 — 3 rows returned (Opportunity#0000000010VJC7AS6GQAR6PRZE, Opportunity#0000000010DRH1Q3QYN9WYH4QF, Opportunity#0000000010NMCAB55DP4PWDY6D); (3) records where last_date_to_apply precedes created_at — 361 rows returned (6.3% of the dataset).

Interpretation. The fee field contains no impossible negative values. The 3 non-positive duration records are logically invalid for time-bound opportunities. More significantly, 361 records show an application deadline that precedes the opportunity's creation date, a scenario not achievable under normal, correctly sequenced data entry.

Business Insight. The 361-record deadline anomaly represents a material data-integrity issue with direct user-experience implications: an applicant could encounter an opportunity that appears expired at the moment of posting, undermining platform credibility and potentially suppressing conversion. This points to a likely gap in backend validation logic (e.g., no constraint enforcing last_date_to_apply ≥ created_at) rather than isolated manual data-entry error, given the volume involved. The 3 non-positive duration records, while small in number, would still distort duration-based aggregates unless excluded or corrected.

Summary: Logical consistency checks found no records with a negative fee value, confirming the integrity of the fee field in this respect. Three records were identified with a duration of zero or less (Opportunity#0000000010VJC7AS6GQAR6PRZE, Opportunity#0000000010DRH1Q3QYN9WYH4QF, and Opportunity#0000000010NMCAB55DP4PWDY6D), which is not logically valid for a time-bound opportunity. More notably, 361 records (6.3% of the 5,730 total) have a last_date_to_apply that falls before their created_at date — a pattern consistent with a backend validation gap rather than isolated data-entry error, and one that warrants correction before these fields are used in deadline or lifecycle analysis.

Section 14: Correlation Analysis

Observation. Correlation matrix among duration, fee, and microscholarship: duration–fee 0.023751; duration–microscholarship -0.015989; fee–microscholarship 0.240942.

Interpretation. Duration shows negligible correlation with both fee and microscholarship, indicating opportunity length has no meaningful linear relationship with either cost or incentive value. Fee and microscholarship show a weak-to-moderate positive correlation, suggesting a loose tendency for higher-fee opportunities to also carry higher microscholarship values, though the relationship is far from strong.

Business Insight. Duration is not a reliable basis for predicting or explaining an opportunity's cost or scholarship value, and should not be used as a proxy variable for either in reporting or modeling. The modest fee–microscholarship relationship suggests pricing and incentive strategies may be loosely coordinated in practice, but are largely driven by other, unmeasured factors (e.g., opportunity category, sponsoring partner, or content type) that fall outside this dataset's current scope.

Summary: *Correlation analysis among the three numerical variables shows a negligible relationship between duration and fee ($r = 0.0238$) and between duration and microscholarship ($r = -0.0160$), indicating that opportunity length has no meaningful linear relationship with either cost or scholarship value. A weak positive correlation was observed between fee and microscholarship ($r = 0.2409$), suggesting a modest tendency for higher-fee opportunities to be associated with higher microscholarship amounts, though the relationship is not strong enough to be treated as a reliable predictive pattern.*

Section 15: Outlier Analysis (IQR Method)

Observation. Using the $1.5 \times \text{IQR}$ method: duration – 134 outliers (2.3% of 5,721 non-null values); fee – 1,208 outliers (21.1% of 5,726 non-null values); microscholarship – 1,641 outliers (28.7% of 5,725 non-null values).

Interpretation. Duration's outlier proportion is modest and consistent with its earlier-observed moderate skew. Fee and microscholarship both carry substantially higher outlier proportions, consistent with their pronounced right-skew and the wide divergence between their means and medians reported in Section 10.

Business Insight. With over one-fifth of fee records and nearly three-tenths of microscholarship records classified as statistical outliers, both variables reflect genuine tiered structures within the platform's offerings — a standard free-or-low-cost tier alongside a smaller premium tier — rather than data-entry noise alone. This supports a segmented reporting and marketing approach: standard offerings and premium/high-value offerings should be analyzed and communicated separately rather than blended into a single average.

Summary: *Outlier detection using the interquartile range (IQR) method identified 134 outliers in duration (2.3% of 5,721 non-null values), 1,208 outliers in fee (21.1% of 5,726 non-null values), and 1,641 outliers in microscholarship (28.7% of 5,725 non-null values). The substantially higher outlier proportions in fee and microscholarship confirm both variables are shaped by a distinct high-value segment, reinforcing the earlier recommendation that mean-based statistics for these fields be paired with medians and, where possible, tier-segmented reporting.*

Overall EDA Conclusion

The Opportunity Analysis-Ready dataset consists of 5,730 unique records (confirmed by opportunity_id cardinality) across 20 columns, spanning opportunity creation dates from August 8, 2022 to May 28, 2026, with zero duplicate rows. The dataset is led by Internship (1,591), Career (1,087), and Competition

(734) category types, is transacted predominantly in USD (4,698 of 5,387 non-null currency records), and, among the 74.6% of records with available location data, is overwhelmingly delivered in remote formats (Work From Home: 2,356; Virtual: 1,918). Three numerical fields — duration, fee, and microscholarship — show varying degrees of right-skew, with fee and microscholarship both carrying high IQR-based outlier counts (1,208 and 1,641 respectively) and means (57.45 and 175.66) considerably exceeding their medians (0.0 and 120.0).

Data quality concerns identified include: high missingness in `is_archived` (90.84%), `role_responsibility` (81.17%), and `role` (66.96%); inconsistent text casing in the `role` field; templated and placeholder content across the description fields; 3 records with non-positive duration; and 361 records (6.3%) where the application deadline precedes the creation date, pointing to a likely gap in backend validation logic. Correlation analysis found no meaningful relationship between duration and fee (0.0238) or duration and microscholarship (-0.0160), and only a weak positive relationship between fee and microscholarship (0.2409). Collectively, these findings support a dataset that is analytically usable for structural, category-level, and pricing-tier reporting, provided that high-missingness fields are treated as supplementary and known logical inconsistencies are corrected or excluded prior to lifecycle-based analysis.

Key Business Insights

- The dataset contains 5,730 unique opportunity records across 20 columns, with `opportunity_id` confirmed as the sole fully-unique identifier.
- Internship (1,591 records, 27.8%) is the leading opportunity category, followed by Career (1,087, 19.0%) and Competition (734, 12.8%); Program and Xploreu are negligible with 2 records each.
- Among the 74.6% of records with available location data, delivery is almost entirely remote — Work From Home (2,356) and Virtual (1,918) account for 4,274 of 4,275 non-null location records (99.98%); this pattern should be understood as indicative of the available data rather than the full dataset.
- USD is the dominant currency, representing 4,698 of 5,387 non-null currency records (87.2%), followed by INR (355) and EUR (334).
- Duration is most commonly measured in weeks (4,286 of 5,725 non-null records, 74.9%), with a median duration value of 30.0.
- The fee field is highly right-skewed (mean 57.45 vs. median 0.0, std 2,792.12, max 200,000.00), confirming most opportunities are free while a small tier carries substantial fees.
- The microscholarship field shows a similar but less extreme skew (mean 175.66 vs. median 120.0, std 832.73), consistent with a standardized low-value incentive model.

- IQR-based outlier detection found 1,208 outliers in fee (21.1%) and 1,641 outliers in microscholarship (28.7%), compared to only 134 outliers in duration (2.3%), indicating both fee and microscholarship reflect a distinct premium tier rather than random noise.
- Correlation analysis shows only a weak positive relationship between fee and microscholarship (0.2409), and negligible correlations between duration and either fee (0.0238) or microscholarship (-0.0160).
- Opportunity creation volume peaked in 2023 (2,123) and 2024 (2,127), together representing 74.2% of all dated records, before declining to 895 in 2025 and 556 in 2026 (partial year); this trend is indicative and should not be treated as a confirmed measure of business growth or decline.
- Three high-missingness fields — `is_archived` (90.84%), `role_responsibility` (81.17%), and `role` (66.96%) — are unreliable for full-population analysis and should be treated as supplementary.
- The `role` field contains inconsistent casing (e.g., "Manager" 720 vs. "manager" 37; "Tester" 64 vs. "tester" 18), indicating a data-standardization gap that affects 429 unique recorded values.
- There are 0 duplicate rows in the dataset, confirming row-level uniqueness across all 5,730 records.
- Logical inconsistencies were found in 361 records (6.3%) where `last_date_to_apply` precedes `created_at`, and in 3 records with non-positive duration values — both pointing to gaps in backend data-entry validation rather than random error.
- Text fields show evidence of templated and placeholder content: a single `long_description` template accounts for 65.0% of non-null entries, and `short_description`, `long_description`, and `role_responsibility` all have a minimum recorded length of 1 character alongside frequent placeholder values such as "test," "abcd," and "Short Description."

Recommendations

- Exclude or flag the 361 records (6.3% of the dataset) where `last_date_to_apply` precedes `created_at` before using these fields in any deadline, application-window, or lifecycle-based analysis, and raise the underlying validation gap with the platform engineering team.
- Investigate and correct the 3 records with duration values of zero or less before including duration in aggregate calculations.
- Standardize the `role` field's text casing (e.g., consolidate "Manager"/"manager," "Tester"/"tester," "Role"/"role") to enable accurate role-based aggregation across its 429 recorded values.
- Report fee and microscholarship statistics using the median (0.0 and 120.0 respectively) alongside the mean, given the substantial skew and high outlier counts identified via the IQR method; consider tier-segmented reporting (standard vs. premium) for both fields.

- Treat `is_archived` (90.84% missing), `role_responsibility` (81.17% missing), and `role` (66.96% missing) as supplementary rather than core analytical fields, given their low population coverage.
- Audit the description fields for templated and placeholder entries — including the single `long_description` template accounting for 65.0% of non-null records — to improve content differentiation and the reliability of any future text-based or NLP-driven analysis.
- Prioritize analysis and reporting around the dominant categories (Internship, Career, Competition, Course, Event), while treating Program and Xploreu as statistically negligible segments.
- Frame location-based findings around the 74.6% of records with available data rather than the full dataset, and prioritize closing the 25.4% location-data gap before drawing firm geographic-strategy conclusions.
- Use `opportunity_id` exclusively as the unique record identifier in joins, deduplication, or reference operations, since both code and name contain minor duplication.
- When analyzing time trends, account for the partial-year nature of the 2026 data (556 records through May 28, 2026) and avoid attributing the 2023 volume surge to organic growth without independent confirmation of whether it reflects scaling or data migration.

VISUALIZATION REPORT

Visualization Notebook (Google Colab): [🔗 Visualizations_Notebook_Tm14.ipynb](#)

(Implemented using Pandas, NumPy, Matplotlib and Seaborn)

Table X. Summary of Exploratory Data Analysis (EDA) Visualizations

The following table summarizes the visualizations created during the exploratory data analysis. Each visualization highlights a specific aspect of the dataset and presents the key insights obtained from the analysis.

Figure in Section	Visualization	Purpose	Key Insight
2.1	Distribution of Opportunity Durations	To examine the distribution of opportunity durations.	Most opportunities have short durations, with a strong concentration around 30 units. The distribution is positively skewed with a few long-duration outliers.
2.2	Fee Distribution (Histogram & Pie Chart)	To examine the distribution and proportion of free and paid opportunities.	Approximately 79% of opportunities are free, while 21% require a fee. Paid opportunities exhibit a wide range of fee values.
2.3	Microscholarship Distribution (Histogram & Pie Chart)	To analyze the distribution of microscholarship amounts.	Around 65% of opportunities offer the standard microscholarship amount, while the remaining opportunities provide varying reward amounts.
3	Outlier Analysis (Box Plots)	To identify potential outliers in the numeric variables.	Duration, fee, and microscholarship all contain outliers, with fee and microscholarship showing the greatest variability.
4	Correlation Between Numeric Fields	To examine relationships among the numeric variables.	Fee and microscholarship exhibit a weak positive correlation, while duration has little to no correlation with the other numeric variables.

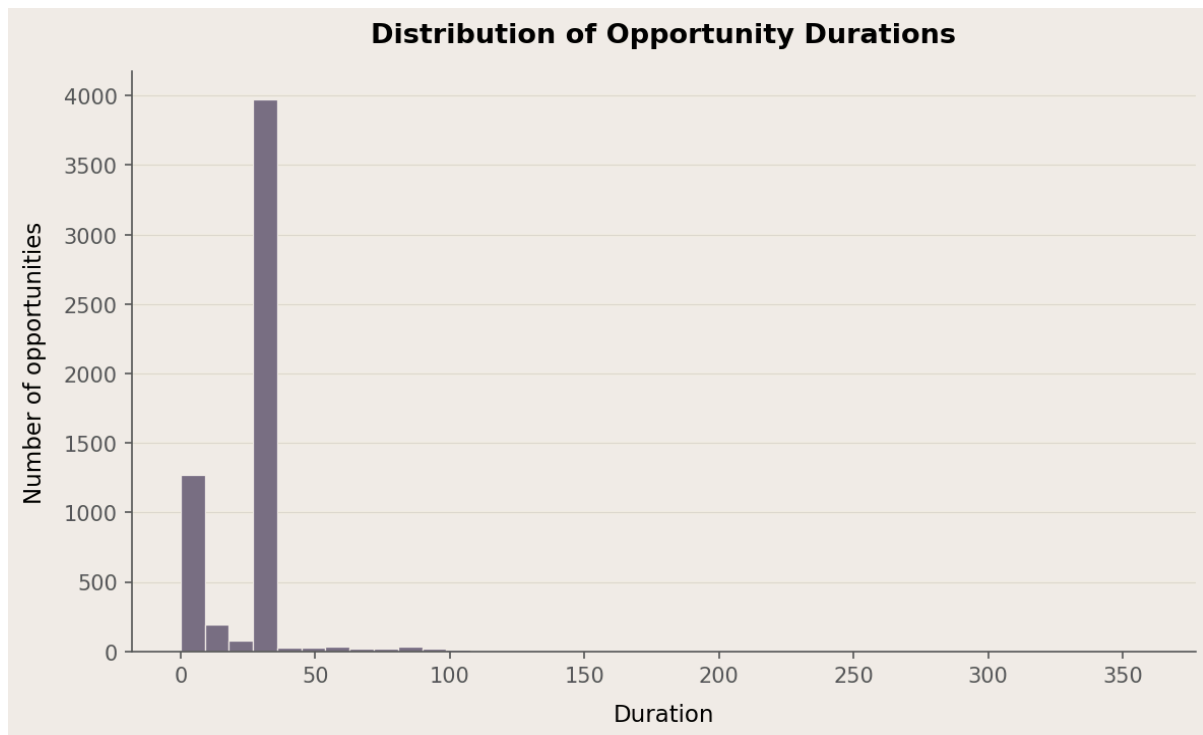
5.1	Opportunity Categories	To identify the most common types of opportunities.	Internships are the most common opportunity type, followed by Career and Competition categories, while Program and XploreU have the fewest opportunities.
5.2	Currencies in Use	To identify the currencies used for paid opportunities.	USD is the primary currency used, with INR and EUR representing only a small proportion of the dataset.
5.3	Duration Units	To analyze the frequency of duration units used.	Weeks are the most frequently used duration unit, followed by months, hours, and days. Minutes and years occur infrequently.
5.4	Opportunity Locations	To examine the distribution of opportunity locations.	Most opportunities are offered as Work From Home or Virtual , indicating a strong preference for remote participation.
5.5	Archive Status	To compare active and archived opportunities.	Archived opportunities significantly outnumber active opportunities, indicating that the dataset primarily contains historical records.
5.6	Approval Type	To analyze the approval process required for opportunities.	Manual approval is the dominant approval method, whereas auto-approved opportunities are comparatively few.

1. DESCRIPTIVE STATISTICS:

	Count	Mean	Std	Min	25%	50%	75%	Max	Missing
Duration	5721	24.5361	16.5902	0	12	30	30	359	9
Fee	5726	57.4549	2792.11	0	0	0	0	200000	4
Micro-scholarship	5725	175.660	832.732	0	100	120	120	50000	5

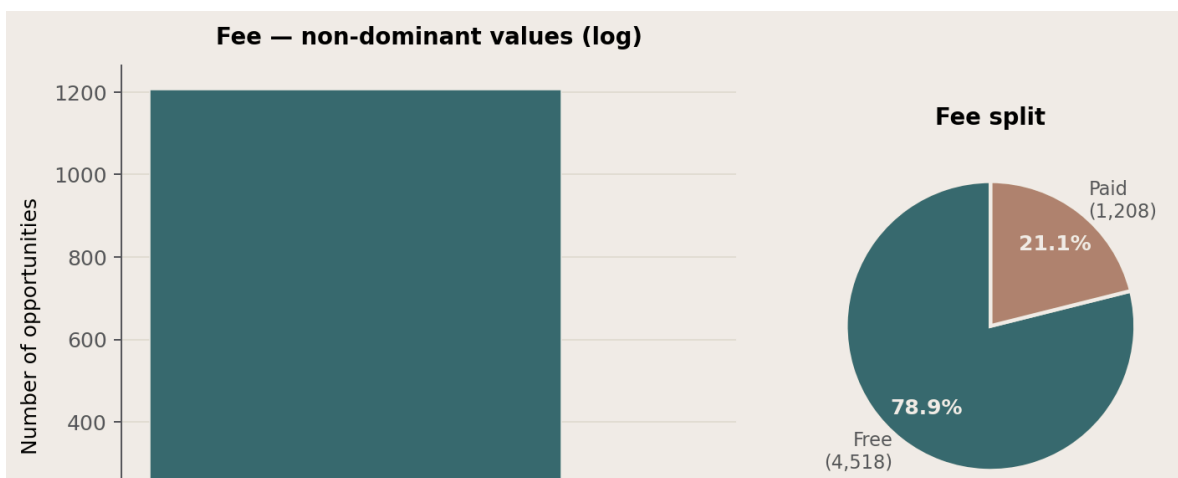
2. NUMERIC VARIABLE DISTRIBUTIONS

2.1: Duration -

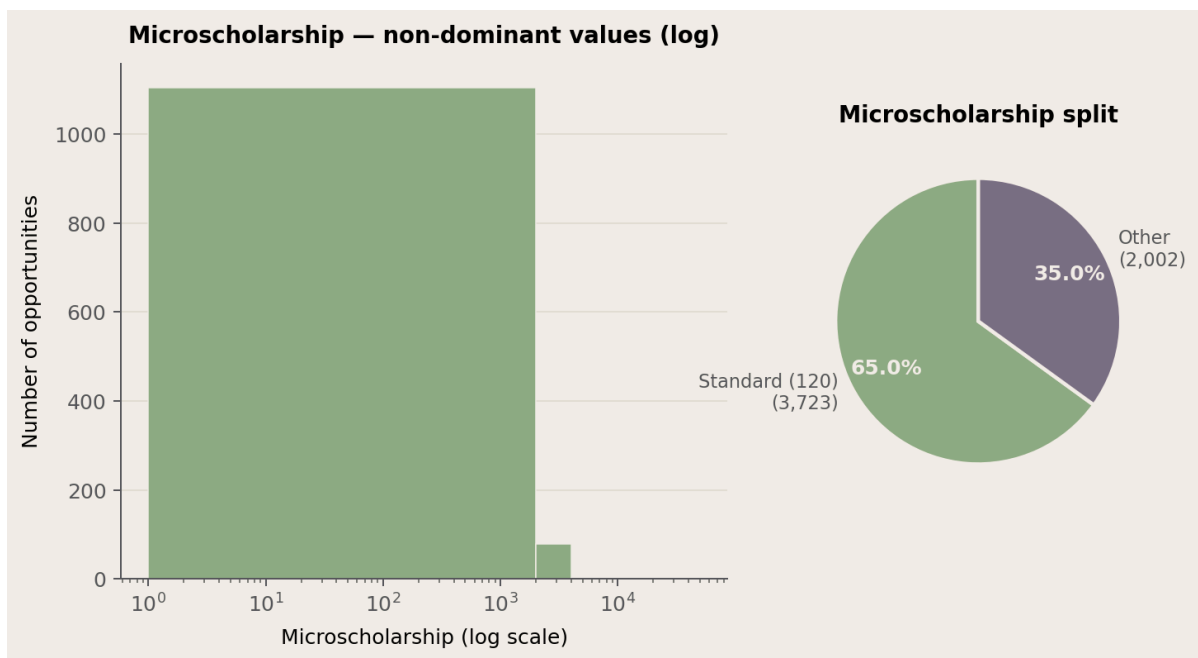


- ❖ This histogram illustrates the distribution of opportunity durations. Most opportunities have short durations, with a strong concentration around 30 units, while only a few extend to much longer durations. This indicates a positively skewed distribution with several long-duration outliers.

2.2: Fee -



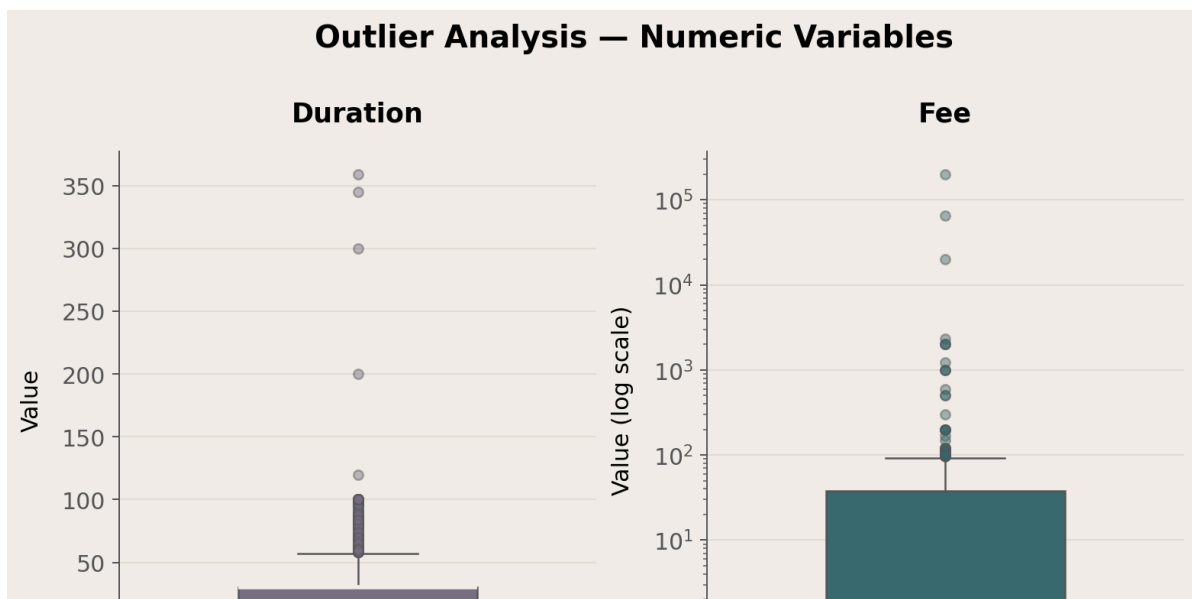
- ❖ The histogram and pie chart together summarize the fee structure of opportunities. Approximately 79% of opportunities are free, while 21% require a fee. Among paid opportunities, fee amounts vary considerably, with a few exceptionally high values. The histogram uses a logarithmic scale because the underlying values are highly skewed.



2.3: Microscholarship -

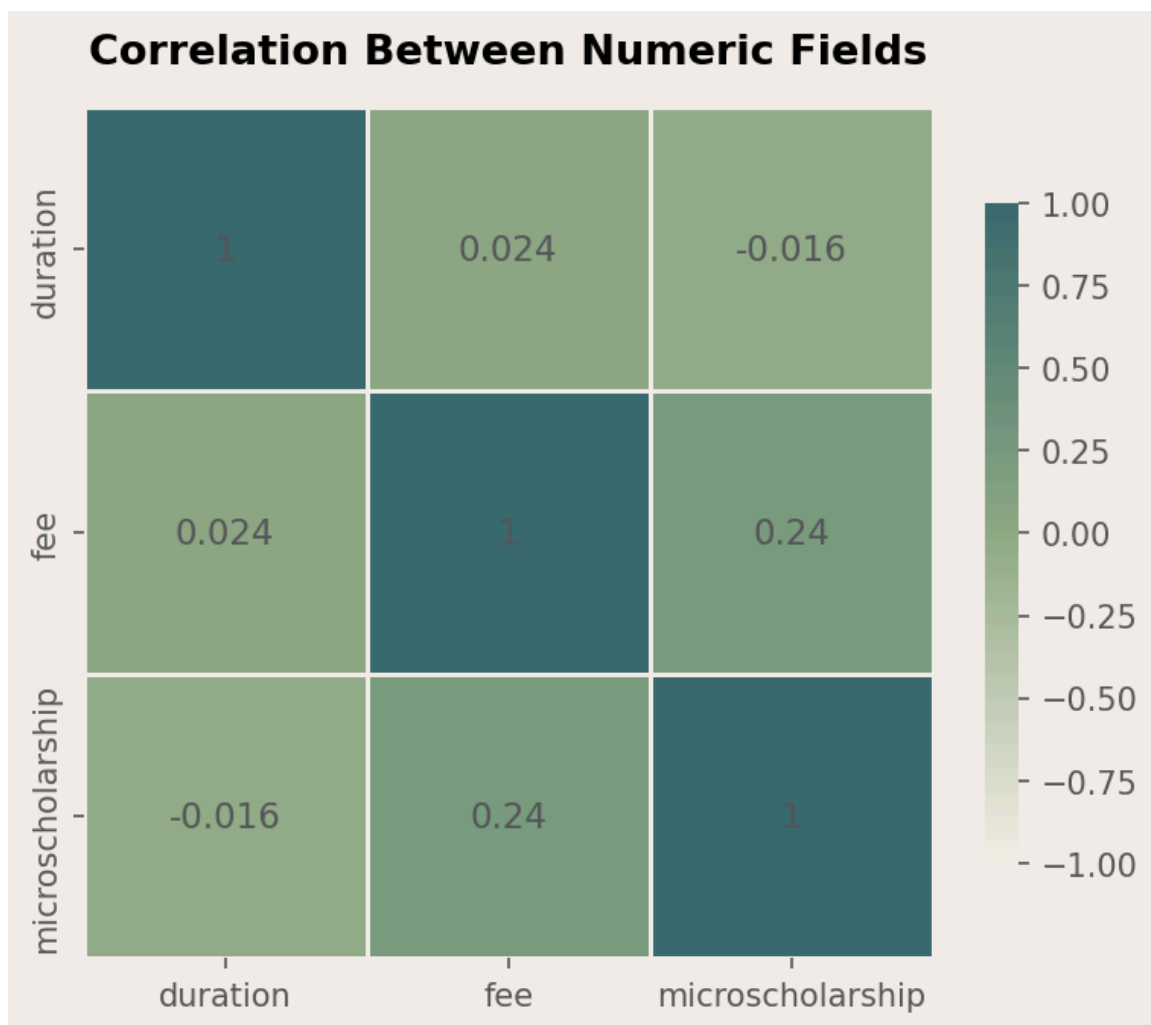
- ❖ This visualization presents the distribution of microscholarship amounts on a logarithmic scale. Around 65% of opportunities offer the standard microscholarship amount of 120, while the remaining opportunities provide different reward values, including a few substantially larger scholarships.

3. OUTLIER ANALYSIS:



- ❖ The box plots identify the presence of outliers in the numeric variables: duration, fee, and microscholarship. All three variables contain outliers, with fee and microscholarship exhibiting the greatest variability due to several extreme values.
- ❖ *NOTE:* "The initial dataset contained extreme values (e.g. a fee of 6 trillion, a microscholarship of 500 million) that appeared inconsistent with plausible business ranges. Following team review, these were treated as invalid entries and set to null before this analysis. The outliers that remain visible in the boxplots reflect moderate values above the typical range and are retained as legitimate business data."

4. RELATIONSHIPS BETWEEN NUMERIC FIELDS



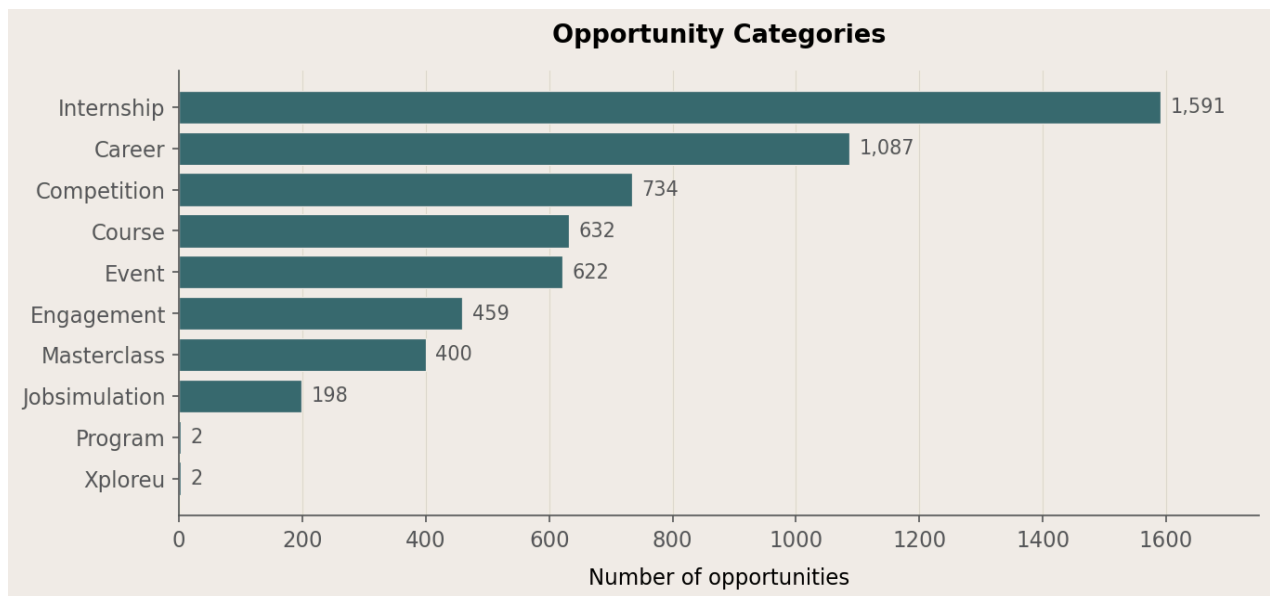
- ❖ The correlation heatmap illustrates the relationships among the numeric variables. Fee and microscholarship show a weak positive correlation, whereas duration has little to

no correlation with either variable. Overall, the numeric variables demonstrate minimal linear relationships.

- ❖ This weak correlation is a useful finding in itself. It suggests the three numeric fields measure independent aspects of an opportunity (financial cost, financial reward, and time commitment) rather than scaling together. In practice, this means opportunities are not "bigger across the board" — a longer opportunity is not necessarily more expensive, and a higher fee does not consistently come with a larger microscholarship. Each variable therefore contributes independent information to any future modelling or analysis.

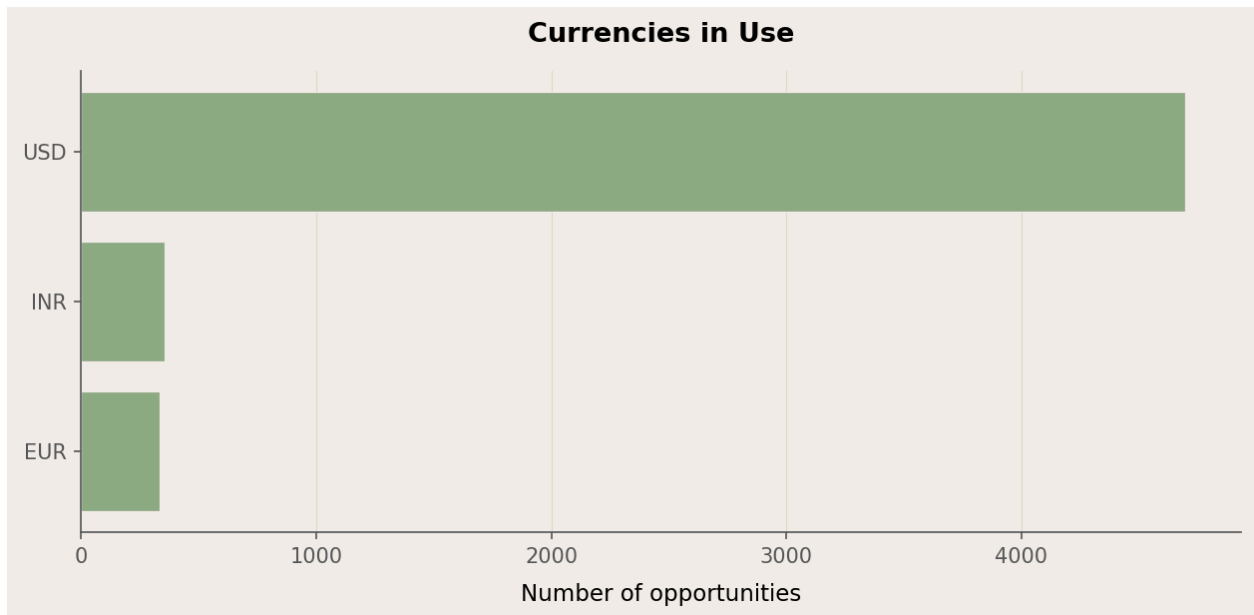
5. CATEGORICAL VARIABLE DISTRIBUTIONS:

5.1: Opportunity Categories -



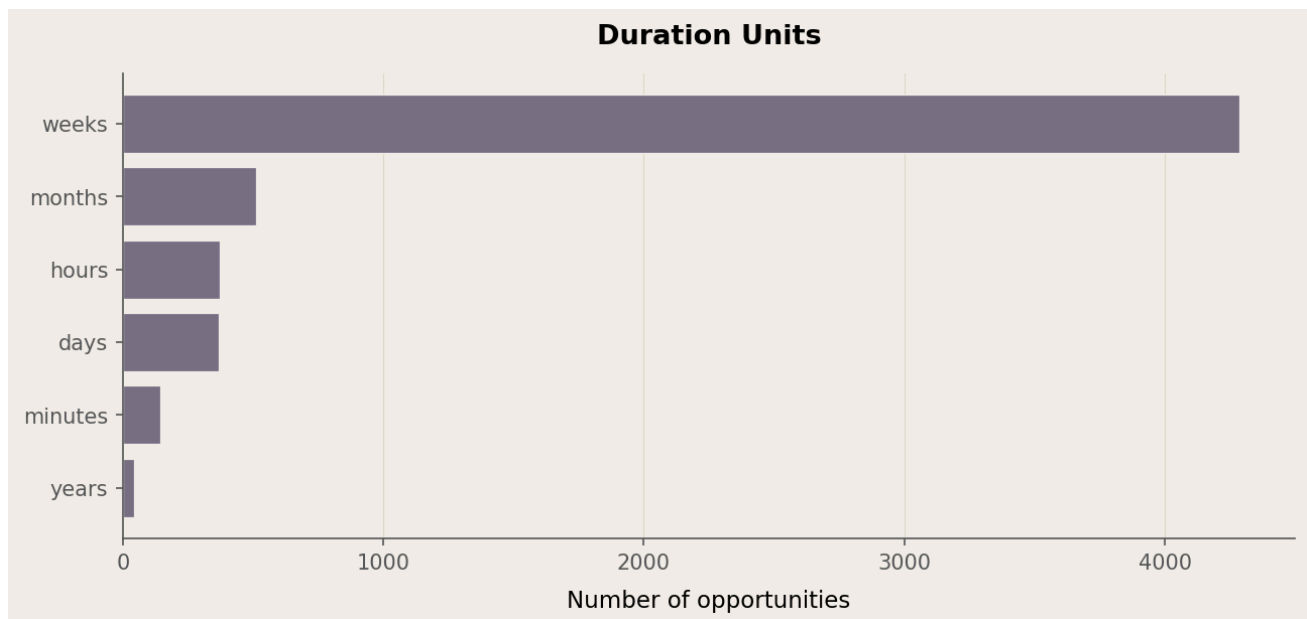
- ❖ This chart presents the distribution of opportunities across different categories. Internships form the largest group, followed by Career, Competition, Course, and Event, which together account for the majority of the dataset. Smaller categories such as Engagement, Masterclass, and Jobsimulation appear less frequently, indicating a platform focused primarily on career development and skill-building.
-

5.2: Currency Distribution -



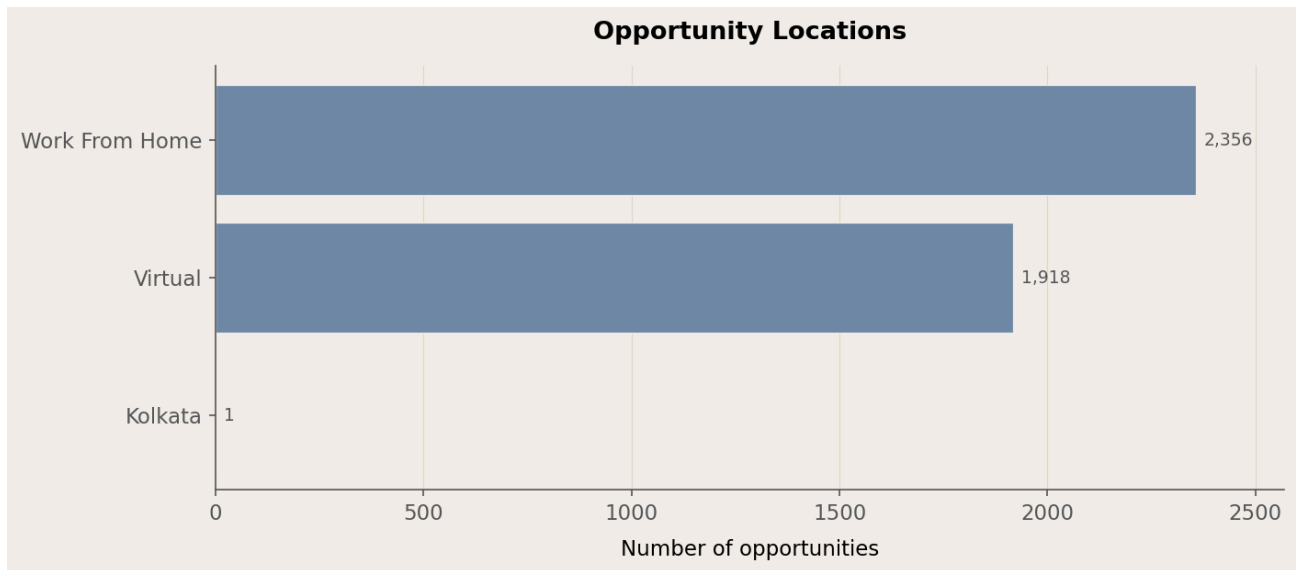
- ❖ This bar chart illustrates the currencies used for paid opportunities. The majority of opportunities are listed in USD, while INR and EUR account for only a small proportion of the total.

5.3: Duration Units -



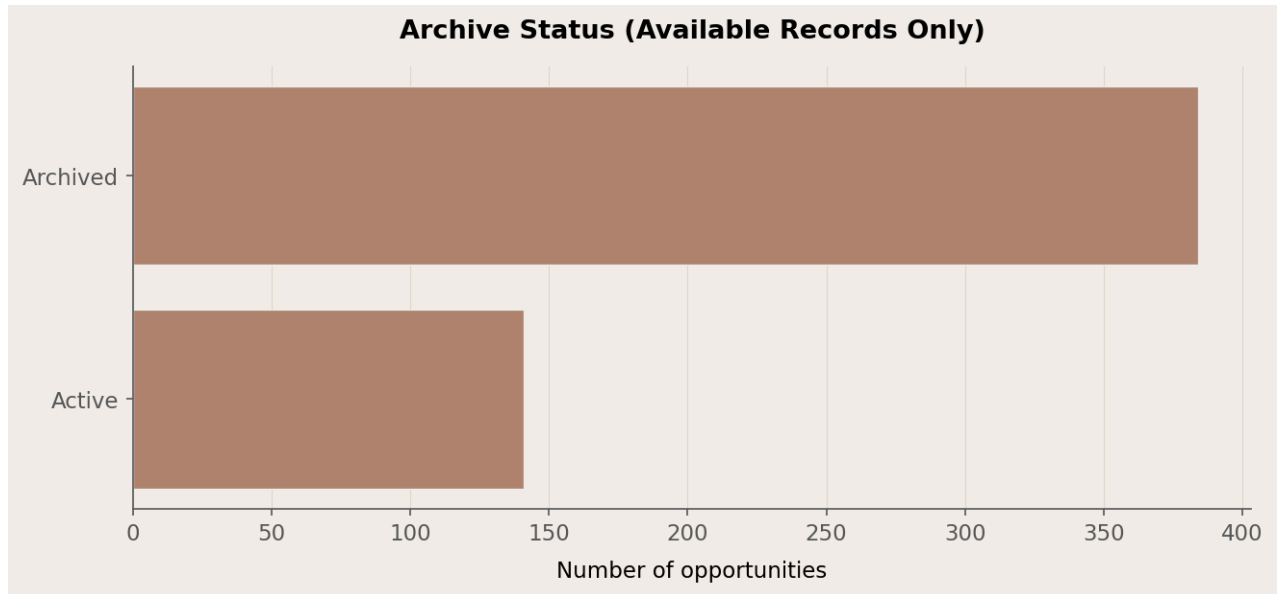
- ❖ This bar chart shows the frequency of different duration units used in the dataset. Weeks are the most commonly used unit, followed by months, hours, and days. Minutes and years are used only in a small number of opportunities.

5.4: Location Distribution -



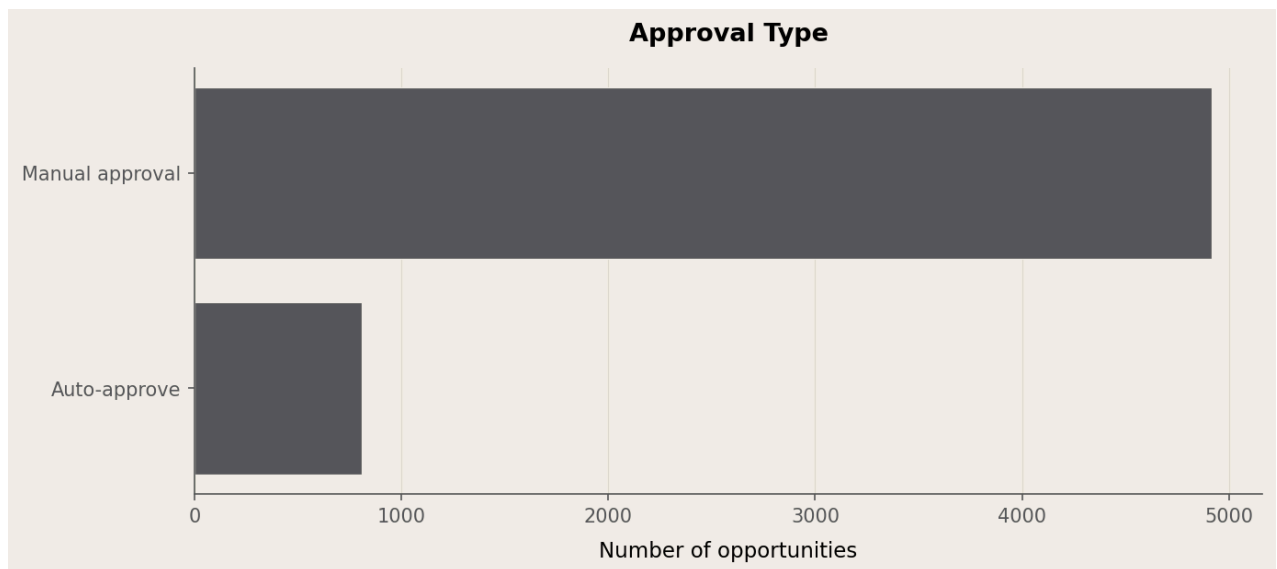
- ❖ This visualization displays the location types of opportunities. The majority are offered as Work From Home or Virtual, indicating a strong emphasis on remote participation, with only one opportunity listing a specific city (Kolkata) and very few requiring a physical location. The column was also cleaned beforehand — placeholder and test entries (such as Latin filler text) were treated as null, and variant spellings like "WFM" and "Work From Home" were consolidated into single canonical labels.

5.5: Archive Status -



- ❖ This chart compares active and archived opportunities within the small subset of records that have a known archive status (approximately 9% of the dataset). Among those records, archived opportunities significantly outnumber active ones. The archive flag is absent for the vast majority of records, so this pattern applies only to the recorded subset.

5.6: Approval Type -



- ❖ This visualization shows the approval process for opportunities. Manual approval is the predominant method, whereas only a limited number of opportunities are automatically approved.

6. Key Observations:

- Numeric distributions are highly concentrated. Roughly 79% of opportunities are free, about 65% of microscholarships sit at the standard value of 120, and most durations fall between 12 and 30 units.
- Numeric fields are largely independent. Weak correlations across fee, microscholarship, and duration indicate that these variables measure separate aspects of an opportunity.
- Internships dominate the category mix, followed by Career, Competition, Course, and Event.
- Location is overwhelmingly remote. Work From Home and Virtual together account for nearly all entries, with only one opportunity listing a specific city.
- The dataset is USD-dominant, with weeks as the most common duration unit and manual approval as the standard for most opportunities.

Business implications and recommendations arising from these observations are discussed in the following section.

Key Findings, Business Insights & Recommendations

5.1 Key Findings

1. Internship and Career Opportunities Lead the Platform

The analysis of 5,730 unique records shows that **Internships** constitute the largest segment at 27.8%, with **Career** (19.0%) and **Competition** (12.8%) following. Combined, these three categories represent nearly 60% of all listings, highlighting the platform's primary function in supporting career growth and professional development.

2. Virtual Delivery is the Primary Model

Within the subset of records containing location data, 4,274 out of 4,275 entries are classified as either **Work From Home** or **Virtual**. This demonstrates that the platform is almost exclusively focused on remote access. Note that since 25.4% of records lack location details, this trend should be interpreted with some care.

3. Typical Opportunities Incur No Cost

The median fee remains at 0, indicating that the majority of offerings are free. While a small number of premium listings feature fees reaching 200,000, these outliers create a right-skewed distribution. Consequently, the median provides a more representative view of cost than the mean.

4. Incentives Follow Standardized Values

Microscholarship awards are largely concentrated around a median value of 120. The lack of varied high-value awards suggests that the organization utilizes a standardized structure for incentives rather than individualized allocations.

5. Strongest Growth Occurred in 2023 and 2024

The volume of new listings peaked during 2023 and 2024, accounting for 74.2% of all dated entries. While figures appear lower for 2025 and 2026, this is likely because the 2026 data only captures a portion of the calendar year.

6. Data Quality Requires Management Attention

While no duplicates were found, several key variables suffer from high proportions of missing information:

- **is_archived**: 90.84% missing
- **role_responsibility**: 81.17% missing
- **role**: 66.96% missing
- **location**: 25.39% missing

These gaps restrict the depth of geographical and role-based analysis, highlighting a clear need for improved data collection protocols.

7. Inconsistent Content Quality in Descriptions

Evidence suggests heavy reliance on templates, with one specific description appearing in 65% of records. Furthermore, the presence of placeholder text like "test" and "abcd" indicates that content quality is not yet standardized across the platform.

8. Logical Discrepancies in Key Dates

The analysis flagged 361 instances where application deadlines preceded the creation dates, along with three records showing invalid non-positive durations. These logical errors suggest that automated validation rules should be implemented during entry.

9. Numerical Features Show Little Interdependence

Correlation results reveal the following:

- Minimal relationship between duration and fee ($r \approx 0.024$)
- Negligible link between duration and scholarship value ($r \approx -0.016$)
- Weak positive connection between fee and scholarship ($r \approx 0.241$)

This suggests that the length of a program does not influence its pricing or the incentives offered.

10. Identification of High-Value Segments

Using the IQR method, the following outliers were identified:

- 134 duration outliers
- 1,208 fee outliers
- 1,641 microscholarship outliers

This indicates a distinct premium catalog segment that should be reported separately from standard free or low-cost opportunities.

5.2 Business Insights and Recommendations

5.1 Business Insights

The exploratory data analysis revealed several important insights into the Opportunity Analysis dataset. These findings provide a better understanding of the platform's opportunities, user trends, and areas where data quality can be improved.

1. Internship Opportunities Dominate the Platform

Internships represent the largest category within the dataset, accounting for nearly one-third of all opportunities. Career opportunities and competitions are the second and third largest categories, respectively. This indicates that the platform primarily serves students and young professionals seeking employment, career development, and practical work experience rather than purely educational content.

2. The Platform is Predominantly Remote

Almost every opportunity is offered remotely through either **Work From Home** or **Virtual** delivery. Physical locations are almost non-existent, demonstrating that the platform is designed to support global accessibility regardless of geographical location.

3. Most Opportunities are Free

The fee analysis shows that the majority of opportunities require no payment. Although a few opportunities have very high fees, these are exceptional cases that significantly increase the average fee. Therefore, the median fee provides a more accurate representation of the typical opportunity cost.

4. Opportunity Listings Grew Rapidly Between 2023 and 2024

The highest number of opportunities was published during 2023 and 2024, indicating that this period represented the platform's strongest growth. Lower volumes observed during 2025 and 2026 should be interpreted cautiously because the 2026 data only covers part of the year.

5. Data Quality Issues Require Attention

Several variables contain high proportions of missing values, particularly **is_archived**, **role**, and **role_responsibility**. Additionally, inconsistent text formatting, placeholder descriptions, and invalid date

relationships reduce the reliability of certain analyses. These issues highlight the importance of improving data validation during data collection.

6. Opportunity Duration Does Not Influence Cost

Correlation analysis demonstrates that the duration of an opportunity has almost no relationship with its fee or microscholarship value. This suggests that pricing decisions are influenced by other factors rather than the length of the programme.

5.2 Business Recommendations

Based on the findings from the exploratory data analysis, the following recommendations are proposed.

Recommendation 1: Improve Data Validation

Implement automated validation rules to prevent incorrect dates, missing mandatory fields, and invalid duration values from being entered into the system. This will improve data accuracy and reporting reliability.

Recommendation 2: Standardise Data Entry

Standardise text fields such as **Role** by applying consistent naming conventions and capitalization. This will improve reporting accuracy and reduce duplicate categories created by inconsistent text formatting.

Recommendation 3: Use Robust Performance Metrics

Because fee and microscholarship values contain many outliers, future reports should present both the median and mean values. This will provide a more representative view of the data and prevent misleading conclusions.

Recommendation 4: Prioritise High-Quality Fields

Business reporting should primarily rely on well-populated variables such as opportunity category, name, fee, duration, and opportunity ID, while fields with excessive missing values should be treated as supplementary information until data completeness improves.

Recommendation 5: Improve Content Quality

Placeholder descriptions and incomplete text should be removed or replaced with meaningful information. Higher-quality descriptions will improve the user experience and support future applications such as recommendation systems and natural language processing.

Recommendation 6: Focus Business Strategy on Core Opportunity Categories

Since Internship, Career, Competition, Course, and Event opportunities represent the majority of platform activity, business development and marketing efforts should prioritise these categories while monitoring the smaller categories separately.

Recommendation 7: Continue Supporting Remote Opportunities

Given the overwhelming demand for virtual opportunities, the organisation should continue investing in remote programme offerings while exploring opportunities to expand physical or hybrid experiences where appropriate.

Recommendation 8: Use Opportunity ID as the Primary Key

Future reporting, dashboard development, and database integration should always use **opportunity_id** as the primary identifier because it is the only field guaranteed to be unique across all records.

Overall Business Insight

The dataset provides a solid foundation for visualization and descriptive reporting, emphasizing a catalog dominated by remote, career-focused, and free opportunities. However, addressing issues related to data completeness, validation, and text quality is essential for improving the accuracy of future dashboards and organizational decision-making.

Links to Supporting Documents

Original Dataset:  [#OpportunityData](#)

Cleaned Dataset:  [Opportunity_Analysis_Ready.xlsx](#)

EDA Notebook:  [EDA_Opportunity_Dataset.ipynb](#)

Visualization Notebook:  [Visualizations_Notebook_Tm14.ipynb](#)